



Department of Computer Science and Engineering

Course Title: Software Engineering

Code: CSE412

Section: 2

Project Proposal

Project Title: LocalLinker

Submitted To:

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Lecture

Department of Computer Science & Engineering

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1. Introduction

Finding trusted service providers can be difficult. Many people struggle with unreliable workers, unclear pricing, and security concerns. Locallinker solves this by offering a secure platform where customers can easily find, book, and pay for verified professionals.

1.1 Project Overview

Locallinker is an AI-powered Local Service Management System designed to connect customers with verified service providers for home repairs, cleaning, electrical work, and other professional services. The platform ensures a secure and seamless service booking experience by integrating real-time communication, AI-driven recommendations, secure payments, and identity verification. It aims to enhance trust, convenience, and efficiency in local service transactions while providing a structured and fraud-free marketplace.

Requirements as a user story

Customers in need of reliable service providers can register and authenticate themselves through a secure sign-up and login process. Once logged in, they can browse available service providers based on location, ratings, price, and category. After selecting a provider, they can confirm bookings, choose a suitable date and time, and proceed with secure payment using multiple options, including online transactions and OTP verification. Real-time tracking and communication allow customers to interact with providers before service delivery. Once the service is completed, customers can rate their experience, leave feedback, and report any issues, ensuring accountability and continuous platform improvement.

Service providers must register and submit verification documents before listing their services. After admin approval, they can manage bookings, update availability, and receive real-time notifications. Providers benefit from an AI-powered recommendation system that increases visibility based on performance, customer ratings, and demand trends. They can chat with customers, respond to inquiries, and track their service history for better management.

Administrators are responsible for verifying service providers, approving or rejecting applications, and ensuring compliance with platform policies. They monitor system activity, detect fraudulent behavior, and take necessary actions to maintain platform integrity. By generating insights and reports on service trends, user behavior, and platform performance, administrators ensure a secure and efficient service environment.

Locallinker provides a trustworthy, scalable, and user-friendly service booking experience, benefiting customers, service providers, and administrators alike.

1.2 List of Requirements

To ensure a seamless, secure, and efficient service booking experience, the system must fulfill both functional and non-functional requirements. These requirements are derived from stakeholder needs, ensuring the platform is user-friendly, scalable, and secure. Additionally, advanced features, known as wow factors, enhance user engagement and platform reliability.

Functional Requirements (FRs)

1. **User Management** – Secure registration, login, and authentication.
2. **Service Discovery** – Search, filter, and browse service providers.
3. **Booking System** – Schedule, modify, and cancel bookings.
4. **Payment System** – Secure transactions with multiple options and OTP verification.
5. **Real-time Communication** – Integrated chat and notifications.
6. **Admin Oversight** – Provider verification, fraud detection, and compliance monitoring.

Non-Functional Requirements (NFRs)

1. **Security & Privacy** – Data encryption, secure authentication, and fraud detection.
2. **Performance & Scalability** – Optimized for high concurrent user handling and low response time.
3. **Usability & Accessibility** – Intuitive UI, responsive design, and ease of use.
4. **Reliability & Availability** – High uptime, fault tolerance, and continuous system monitoring.

Extraordinary Requirements (Wow Factors)

- **AI-driven Smart Matching** – Recommends service providers based on past ratings and preferences.
- **Live Tracking of Service Providers** – Ensures customer safety and transparency.
- **Real-Life Payment System** – Integrated with real-world payment gateways for seamless transactions.

1.3 Technology Stack:

LocalLinker will be developed using a modern, scalable technology stack that ensures real-time communication, AI-driven recommendations, and secure payment processing. The selected technologies focus on high performance, security, and future scalability. This stack supports modular development, making the platform easy to maintain and upgrade as needed.

The following technologies are proposed:

Category	Technologies
Programming Languages	JavaScript, Python
Frontend Frameworks/Technologies	React.js, Next.js, HTML, CSS
Backend Frameworks	Node.js with Express.js, PHP
Database	MongoDB (NoSQL) or MySQL (SQL)
Authentication	JWT (JSON Web Token), OAuth 2.0
Real-time Communication	WebSocket (or Firebase)
AI Engine	Python libraries (Random Forest, XGBoost)
Deployment & DevOps	GitHub Actions (CI/CD), Docker (Containerization), Cloudflare Workers (Cloud Hosting)
Testing & Development Tools	Jest (Unit Testing), Postman (API Testing)

2. Architectural Design

This section provides both a high-level and detailed view of the LocalLinker system's structure. It illustrates how various system components, users, and external systems interact with one another to deliver the platform's full functionality. The architectural design highlights the system's boundaries, core modules, data flow, and communication pathways. It also explains how internal modules are logically organized to support modular development, scalability, security, and efficient system maintenance as the platform grows.

2.1 Architectural Context Diagram

The Architectural Context Diagram provides a Level 0 overview of the system, representing LocalLinker as a single system block and showing its interactions with primary external actors: Customers, Service Providers, Administrators, and third-party services (e.g., Payment Gateway). It defines the system boundaries and main communication paths.

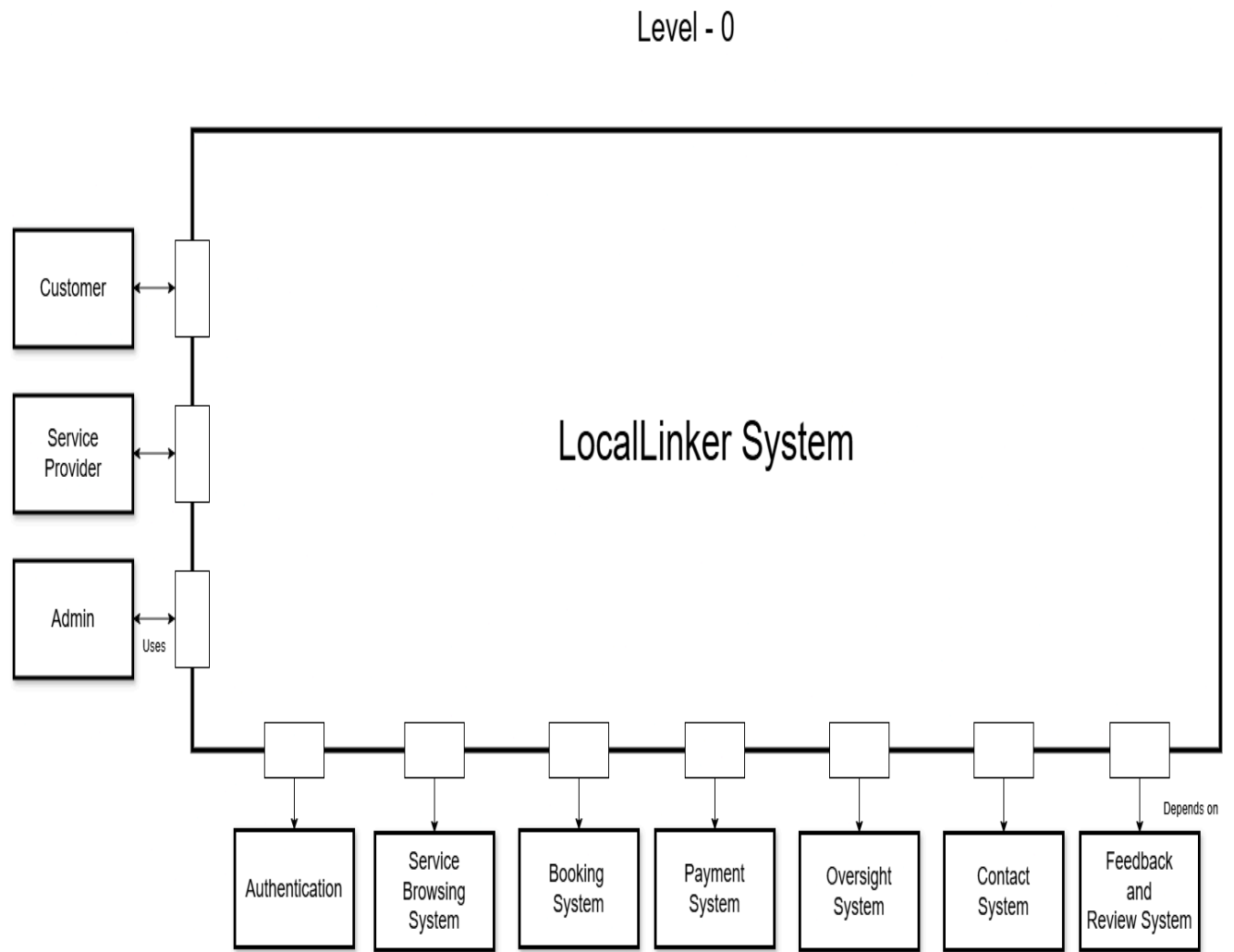


Figure 1: Level 0 for LocalLinker

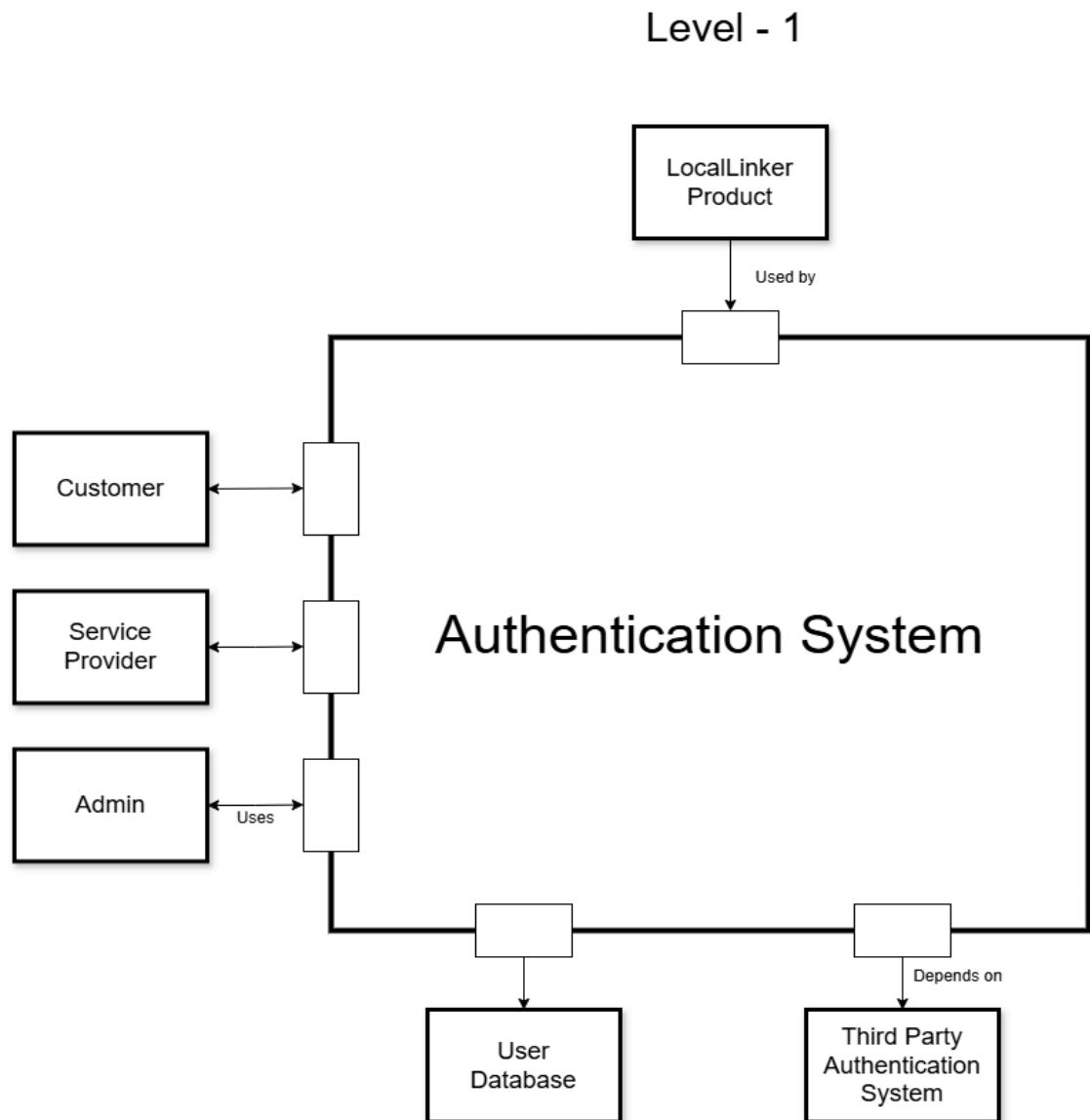


Figure 2: Level 1 for Authentication

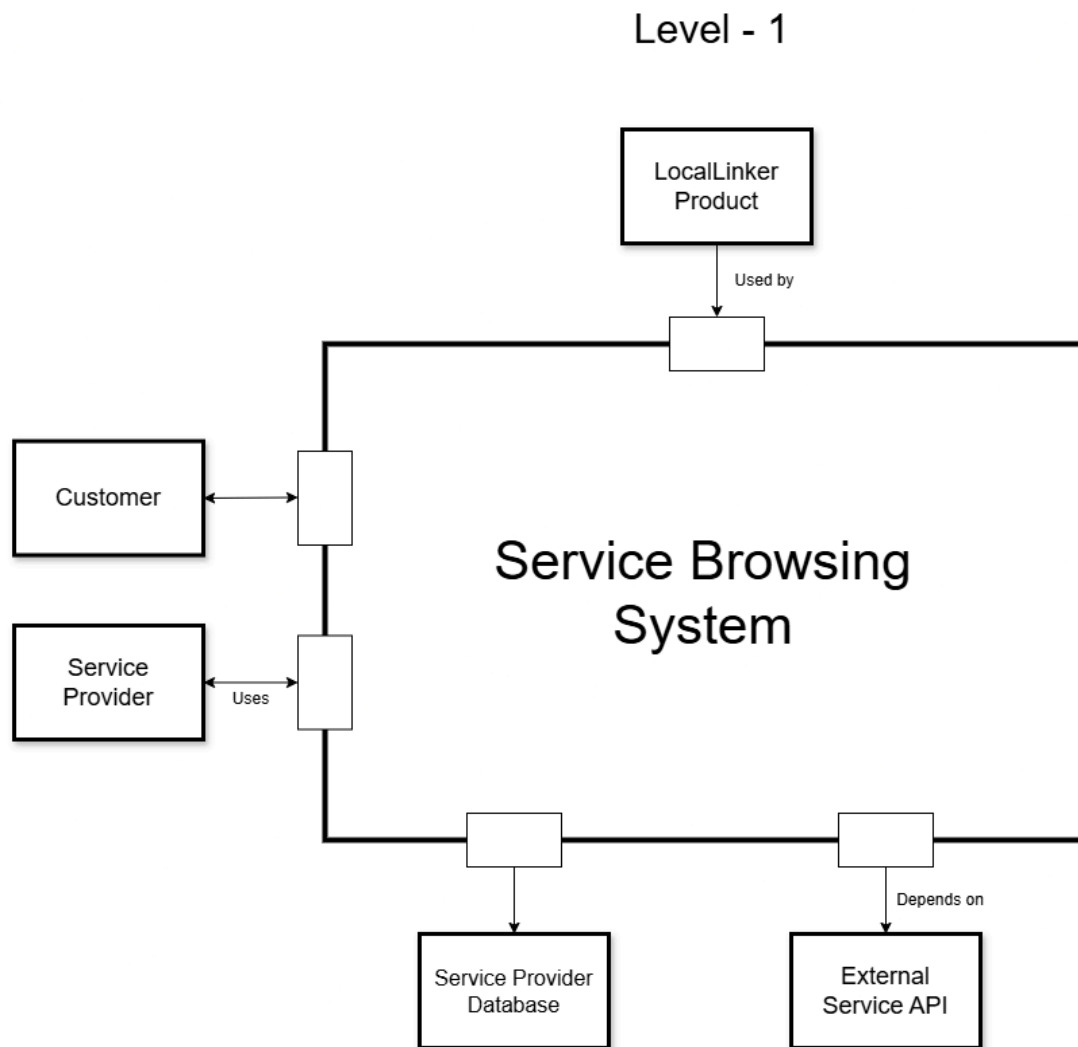


Figure 3: Level 1 for Service Browsing System

Level - 1

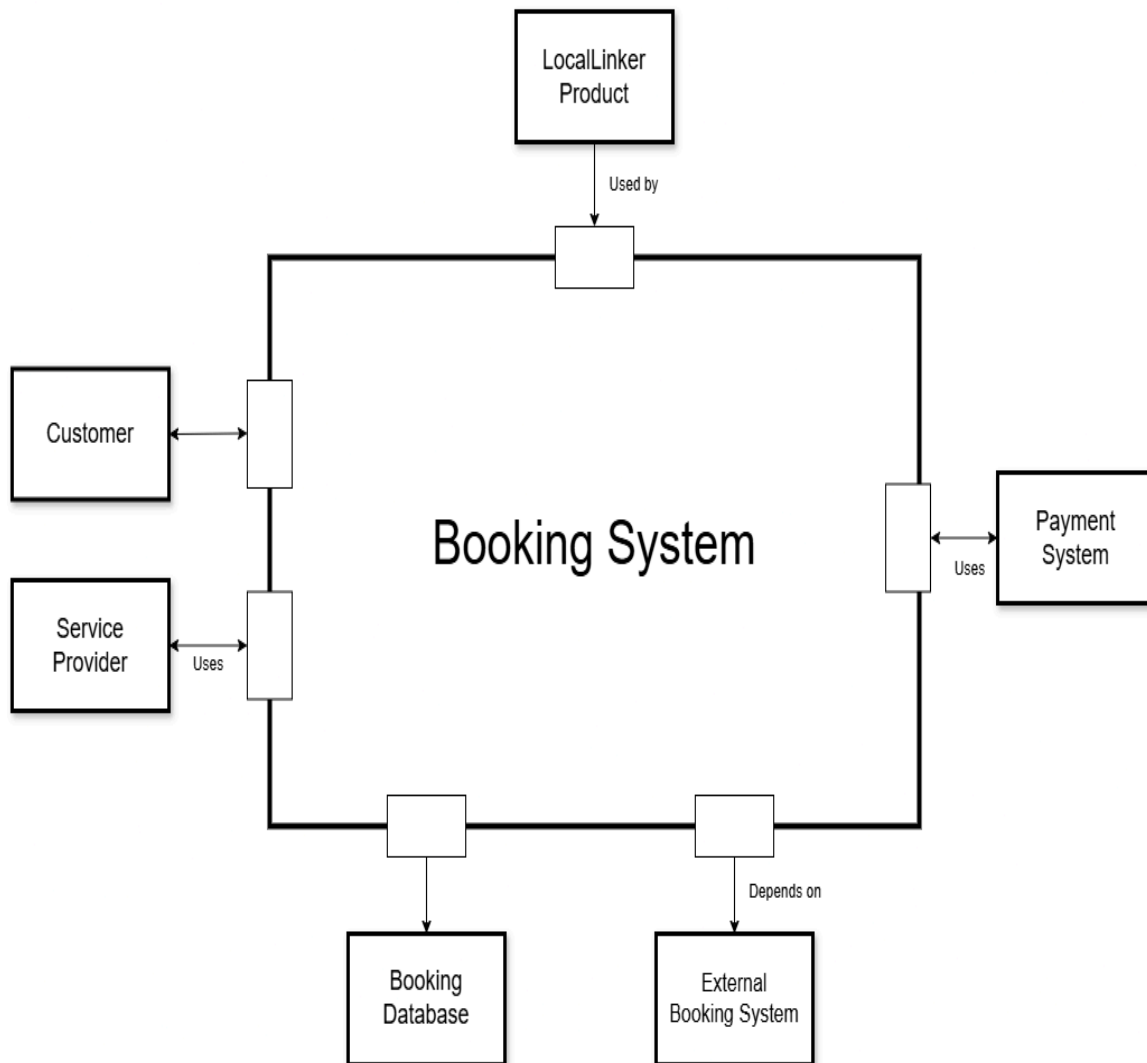


Figure 4: Level 1 for Booking System

Level - 1

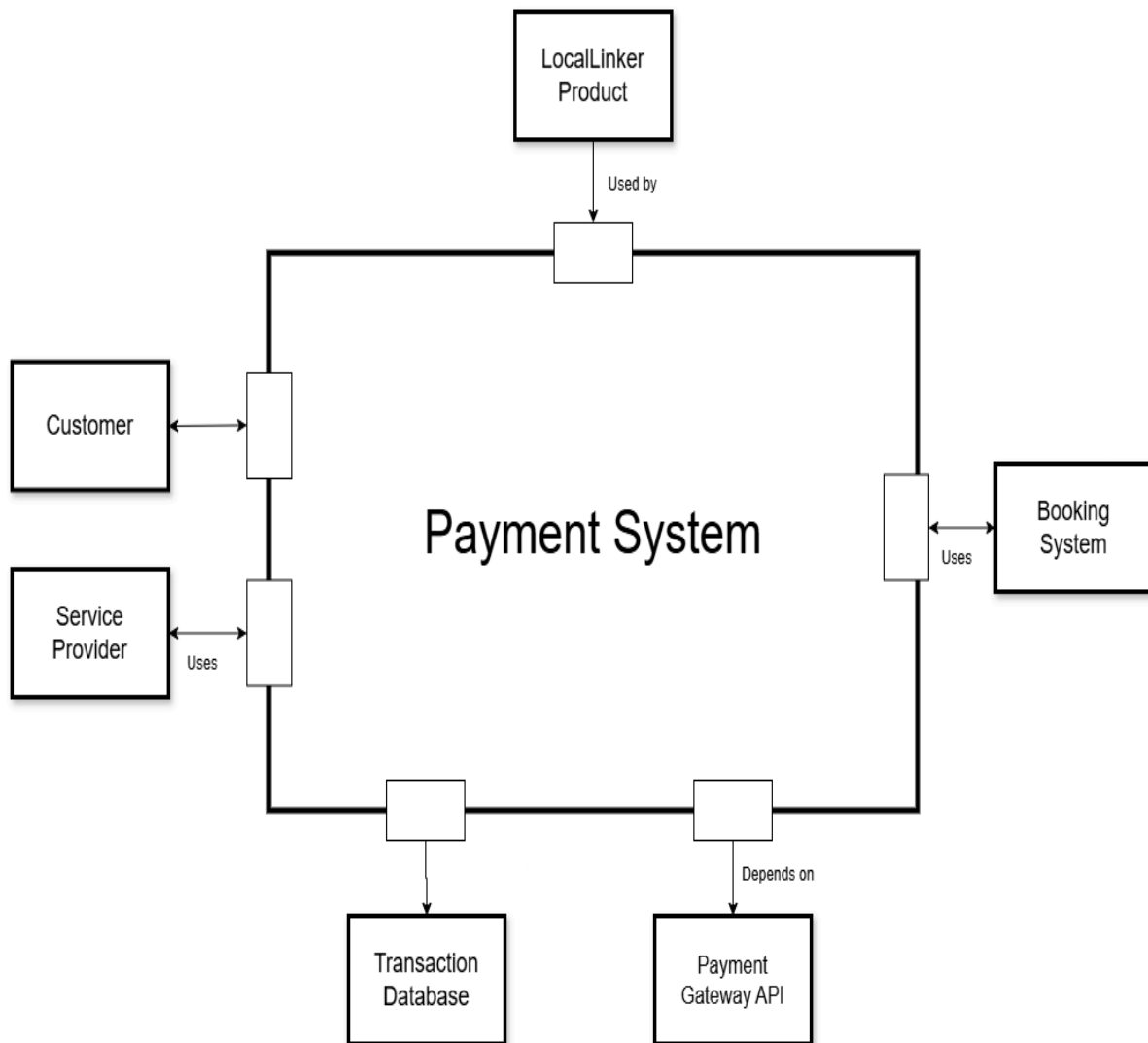


Figure 5: Level 1 for Payment System

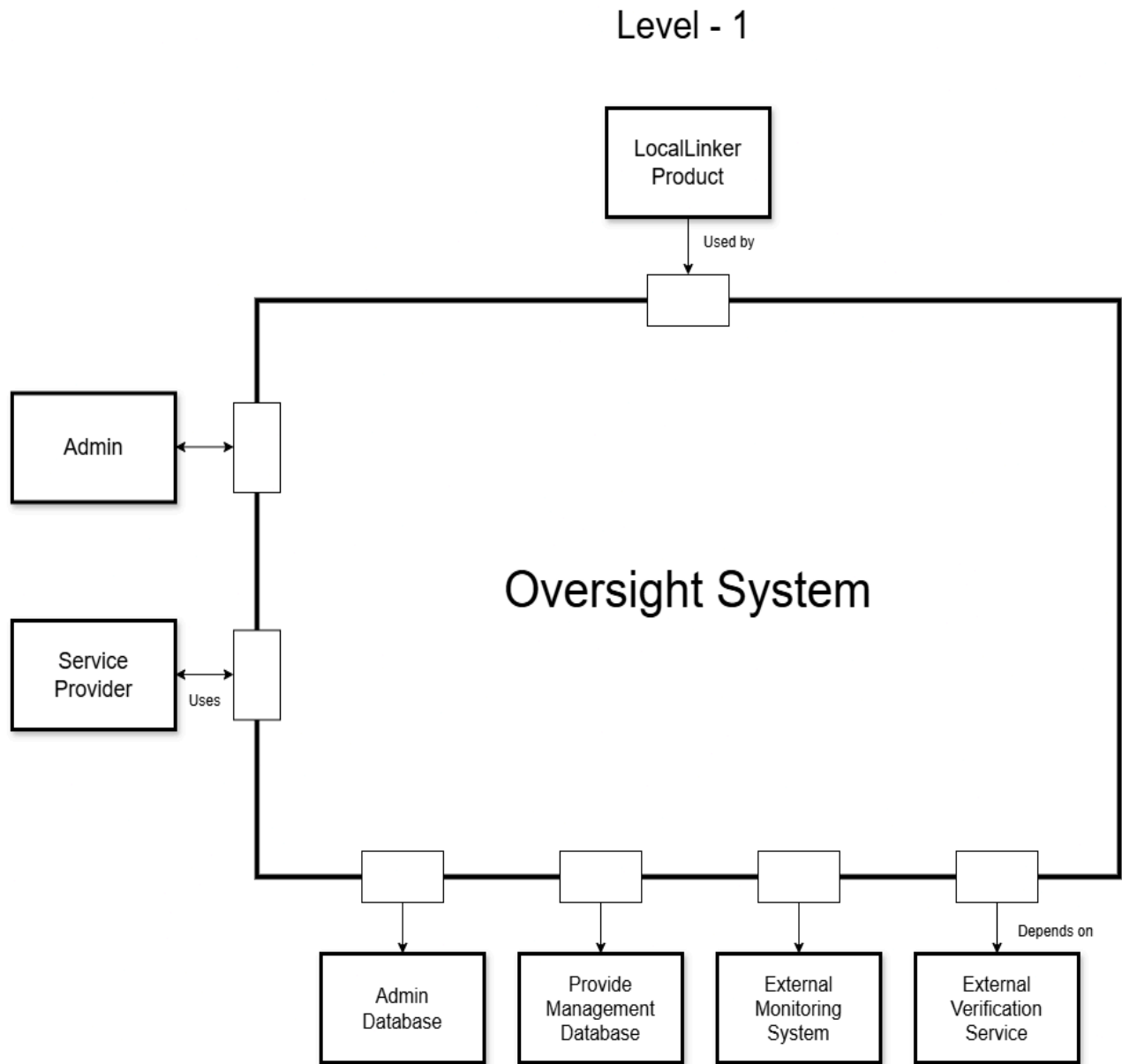


Figure 6: Level 1 for Oversight System

Level - 1

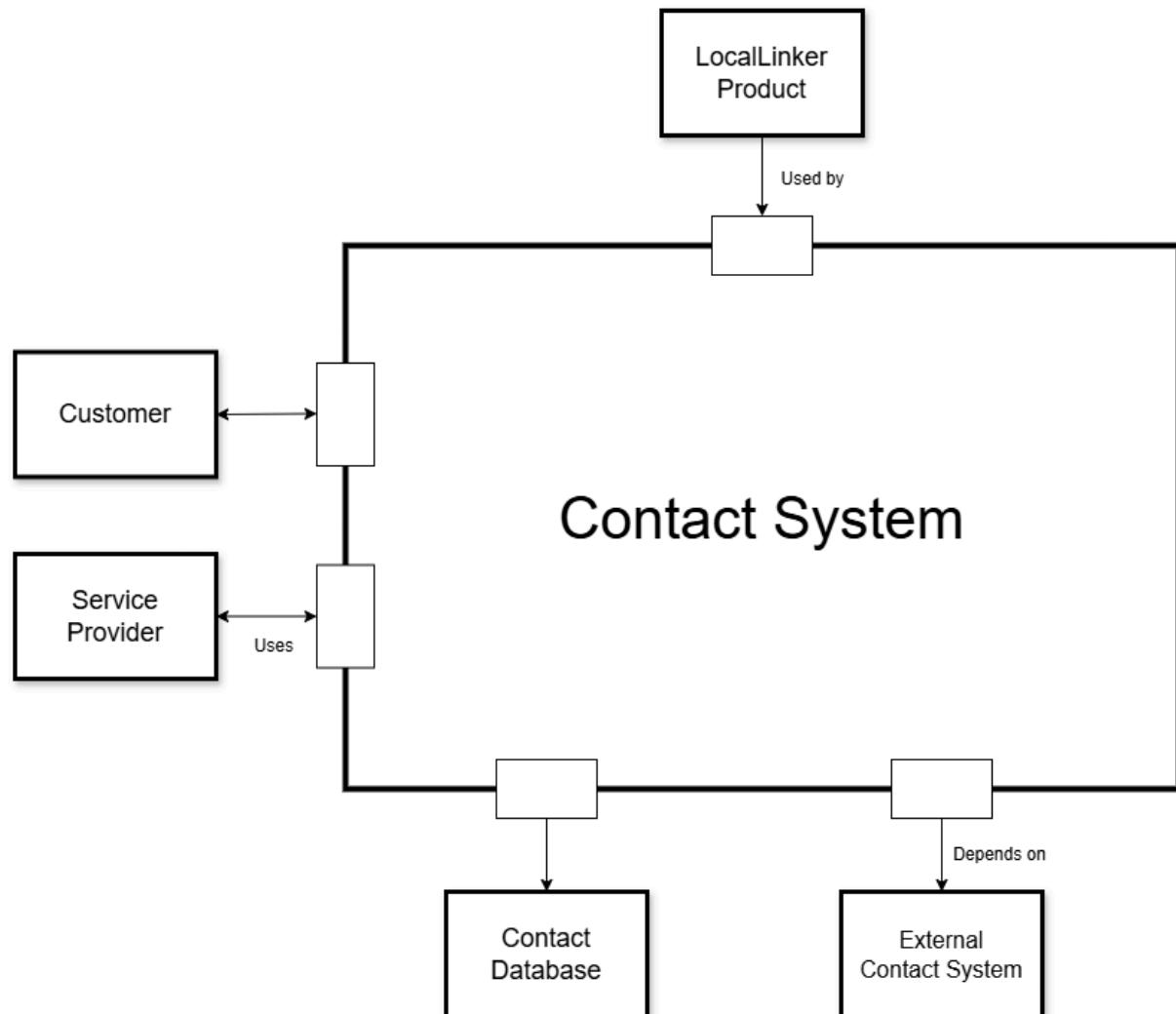


Figure 7: Level 1 for Contact System

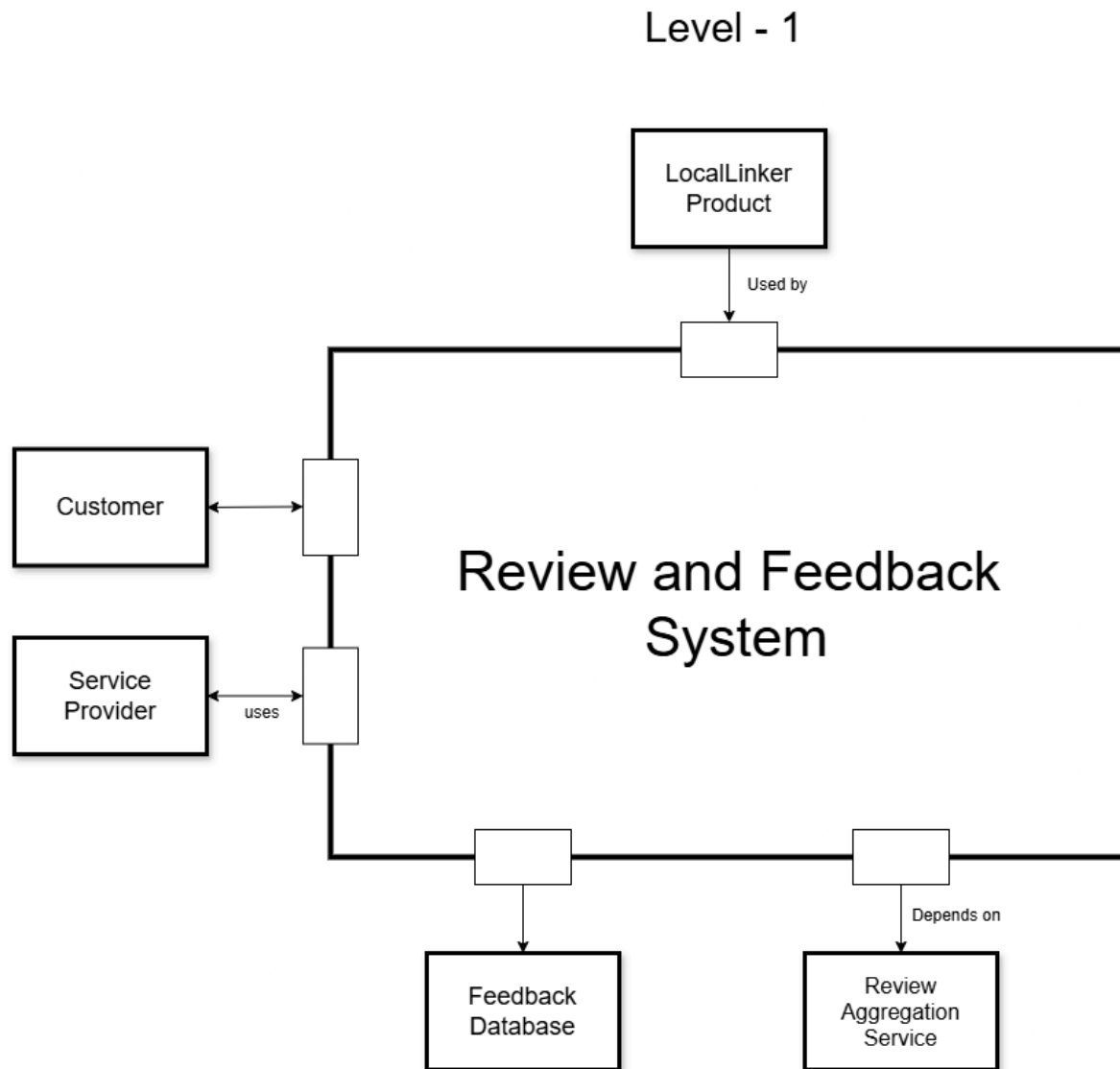


Figure 8: Level 1 for Review and Feedback System

2.2 Top-Level Component Diagram

The Top-Level Component Diagram decomposes the LocalLinker system into major functional modules such as Authentication, Service Management, Booking Management, Payment Processing, Communication Services, and Admin Management. It visually illustrates how these components interact internally to provide complete system functionality.

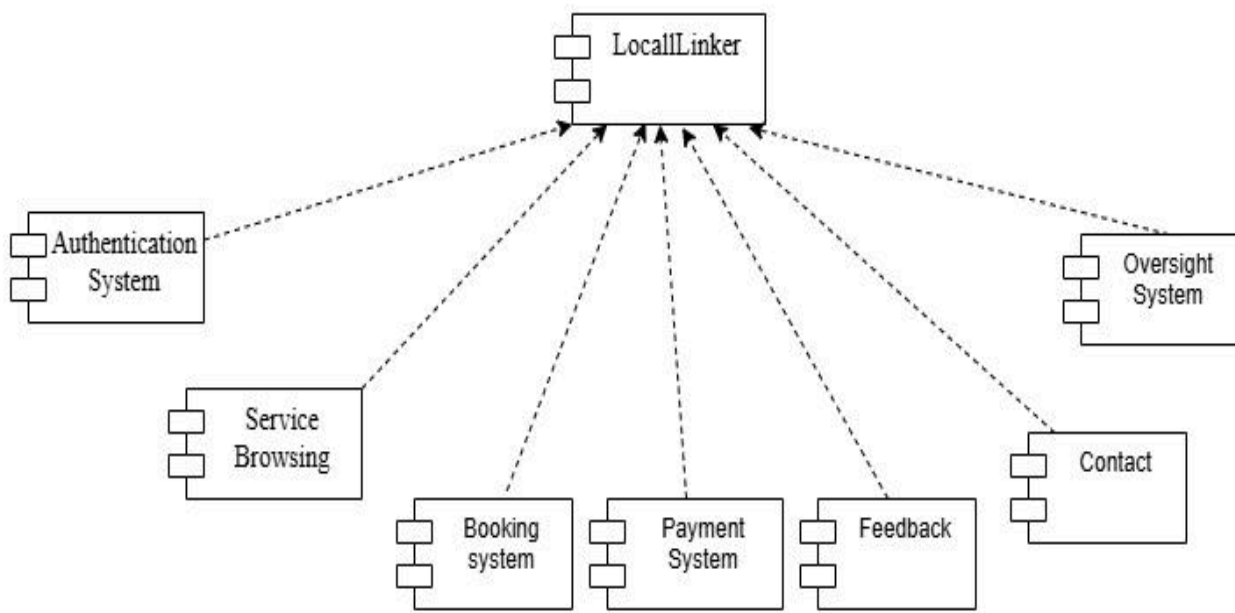


Figure 9:Top-Level Component Diagram for LocalLinker

2.3 Instantiation of Each Component with Component Elaboration

For each top-level component, a detailed elaboration is provided. This includes its internal structure, the responsibilities it handles, how it communicates with other components, and key data it manages. This breakdown ensures modularity, clarity, and ease of future maintenance or scaling.

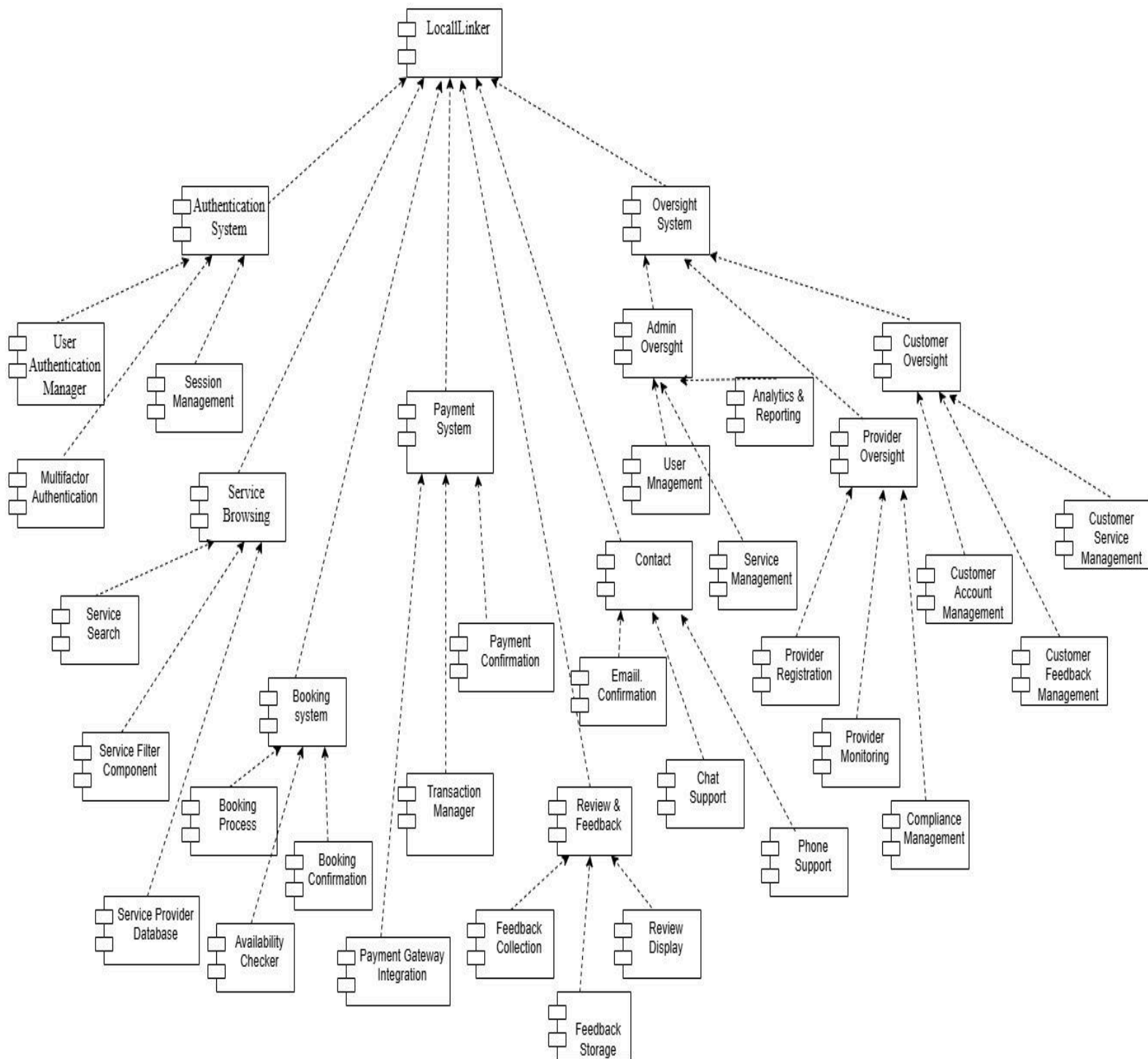


Figure 10:Instantiation of Each Component for LocalLinker

3. Component-Level Design

This section focuses on the detailed design of each major component within the LocalLinker system. It breaks down the internal structure of functional modules, defines key software classes and their interactions, and models how system data will be organized and managed. By elaborating component responsibilities, class structures, and database design, this section ensures that every part of the system is modular, scalable, and maintainable. The objective is to create a robust foundation that simplifies implementation, enhances system flexibility, and supports future upgrades or expansions.

3.1 Elaboration of Design Components

In this phase, each major component is elaborated through a structured approach that includes identifying subsystem responsibilities, modeling the interactions among different modules, and defining preliminary design classes. This detailed breakdown promotes modularity, reusability, and flexibility, ensuring that the LocalLinker system can evolve efficiently as future requirements grow or change.

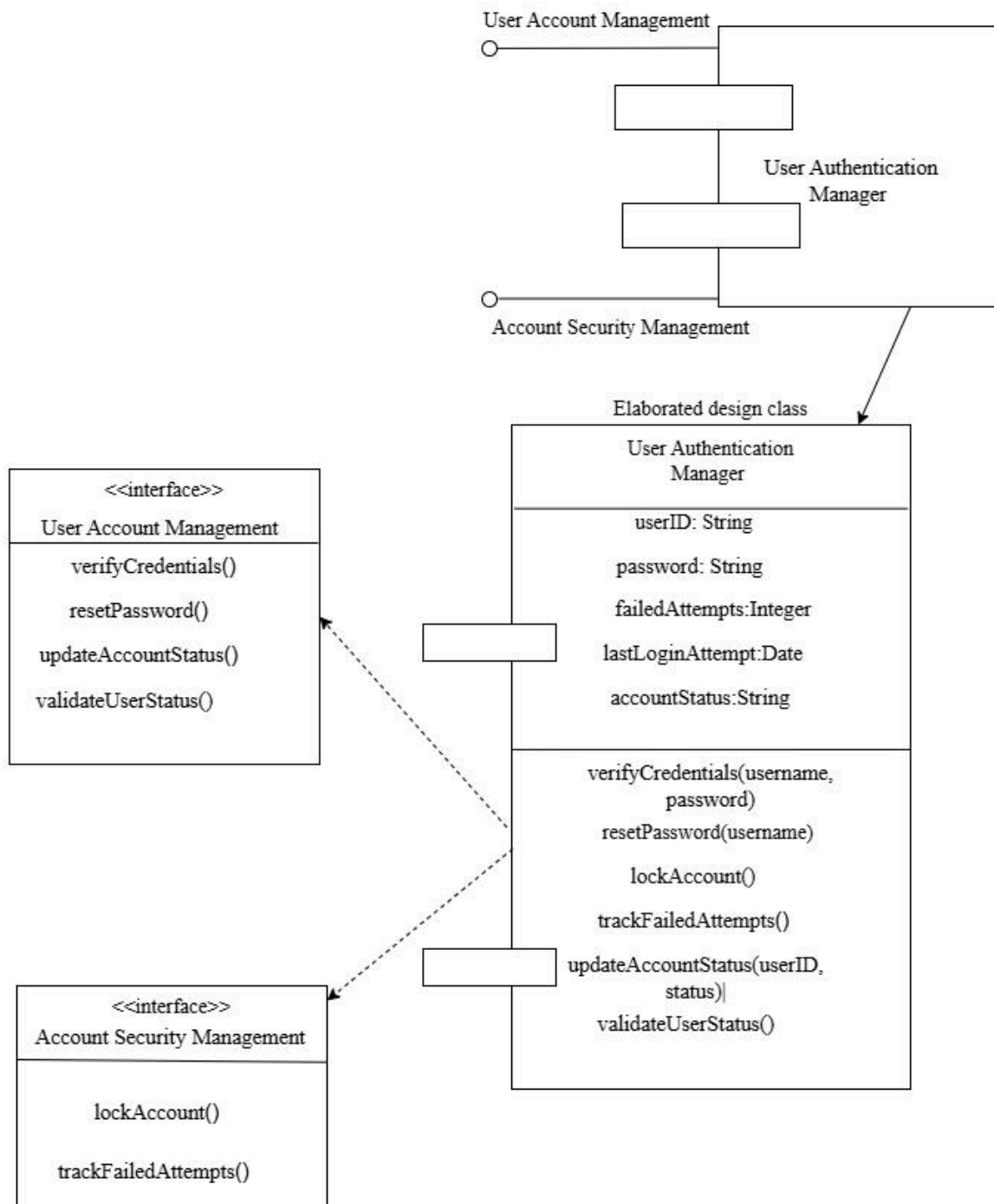


Figure 11:Elaboration of User Authentication Manager

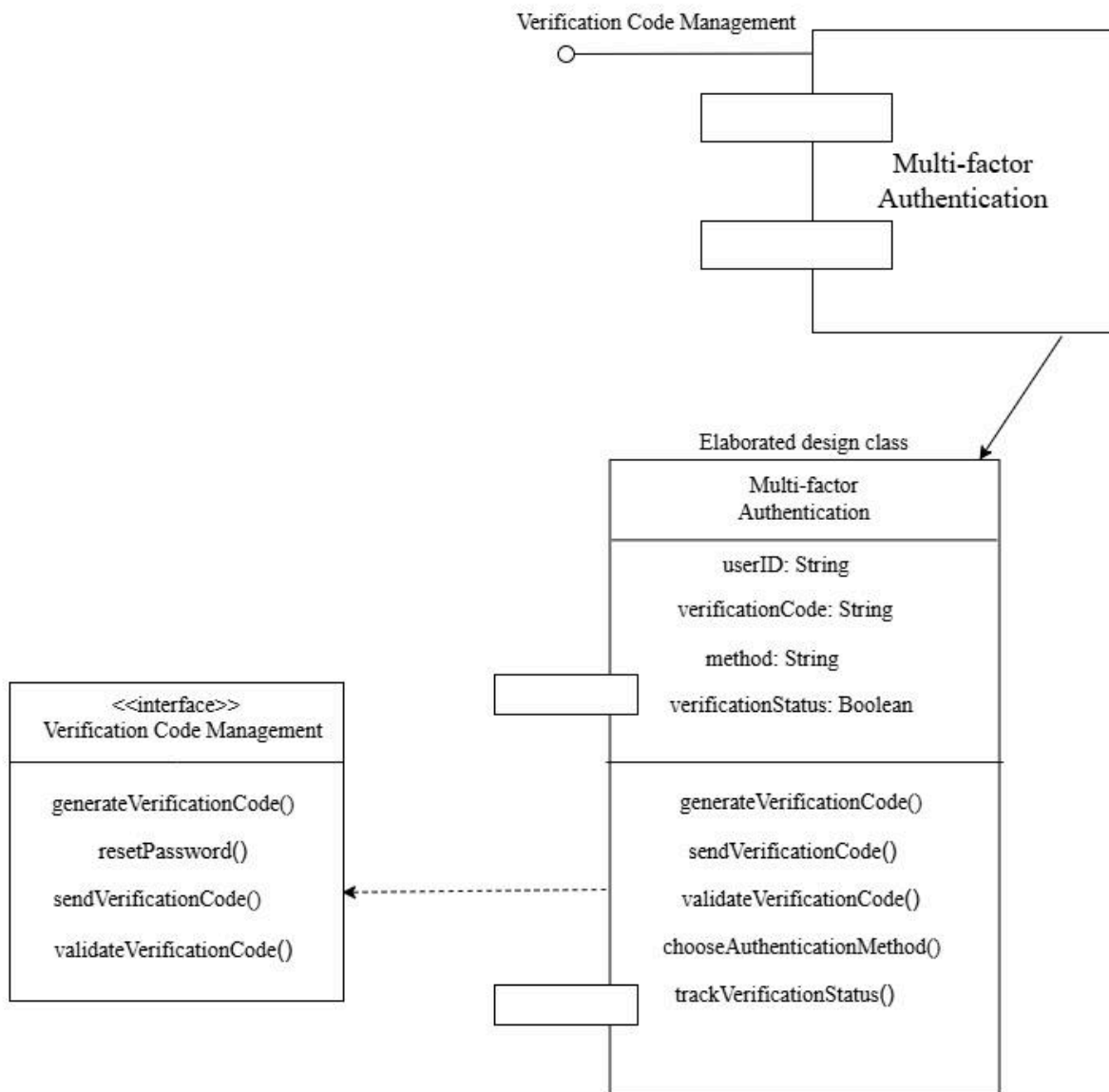


Figure 12:Elaboration of Multi-Factor Authentication

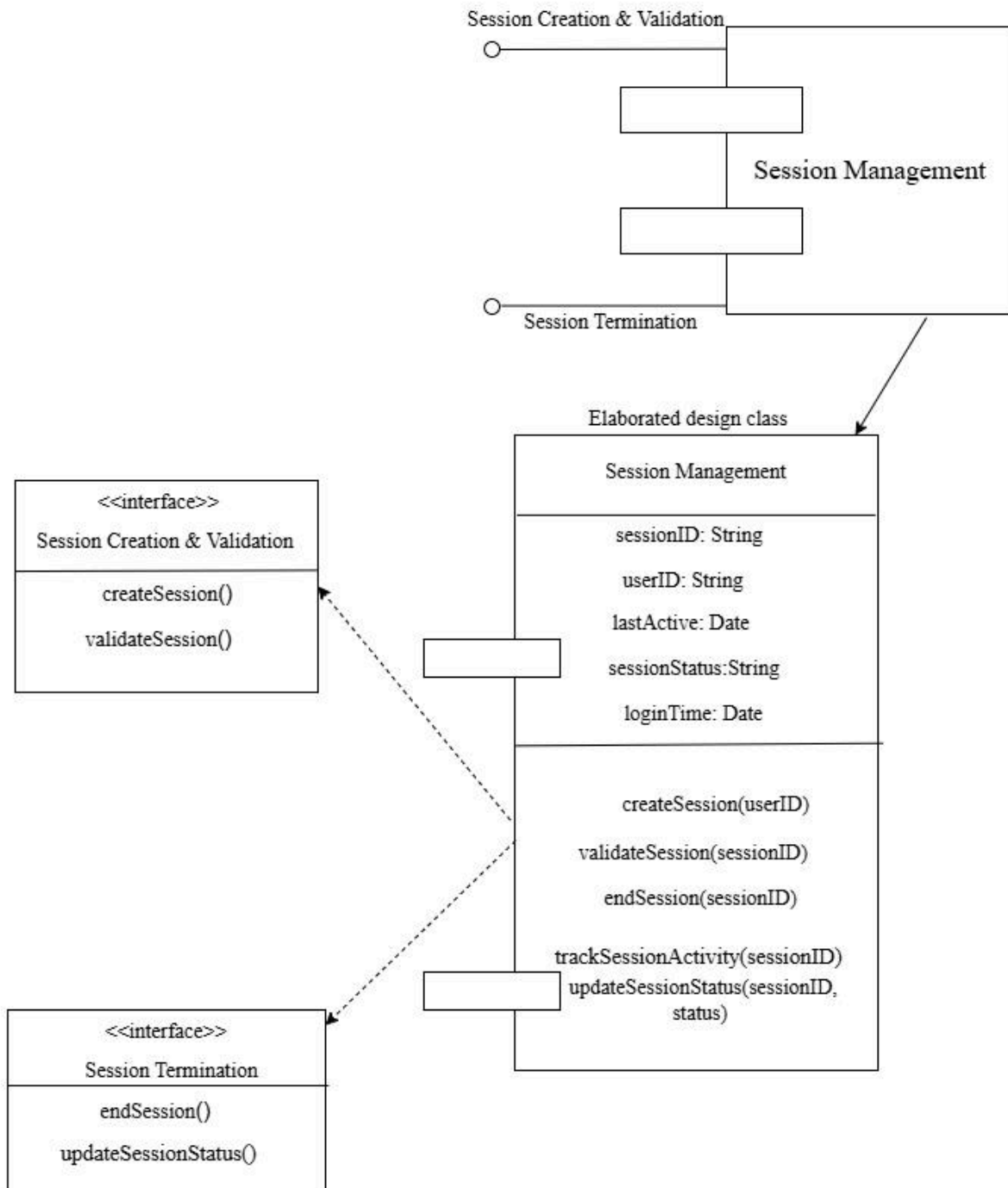


Figure 13:Elaboration of Session Management

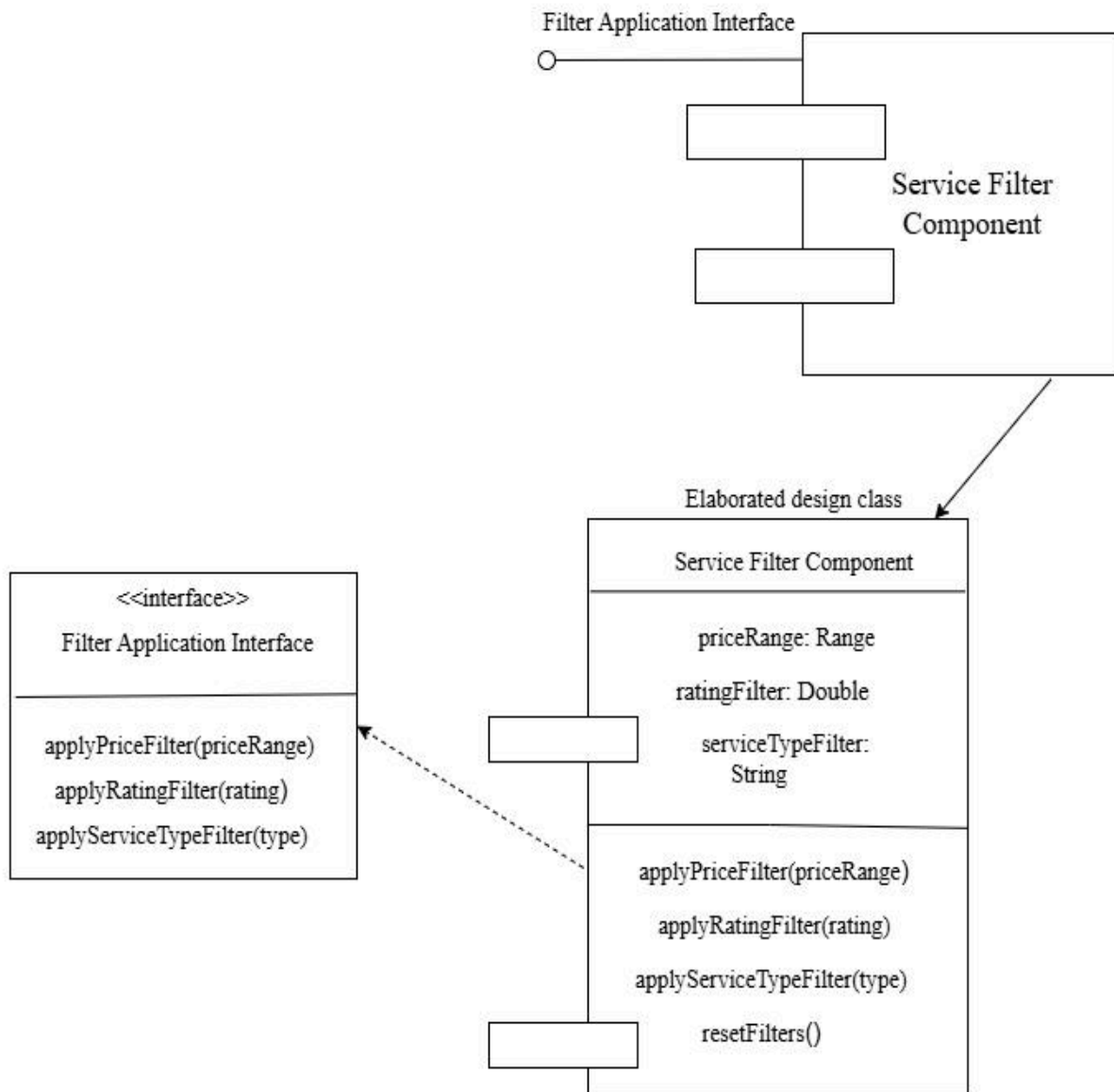


Figure 14:Elaboration of Service Filter Component

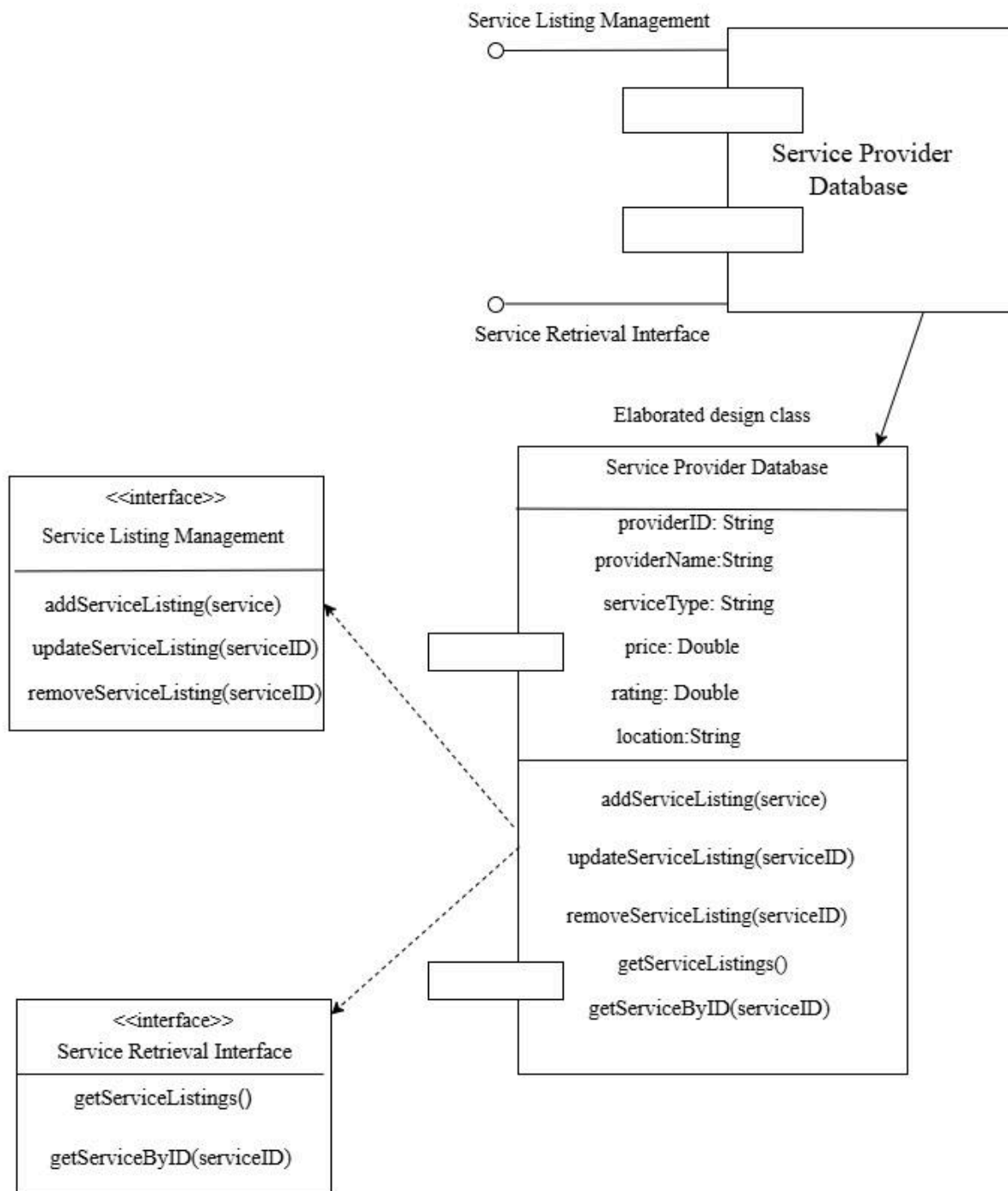


Figure 15:Elaboration of Service Provider Database

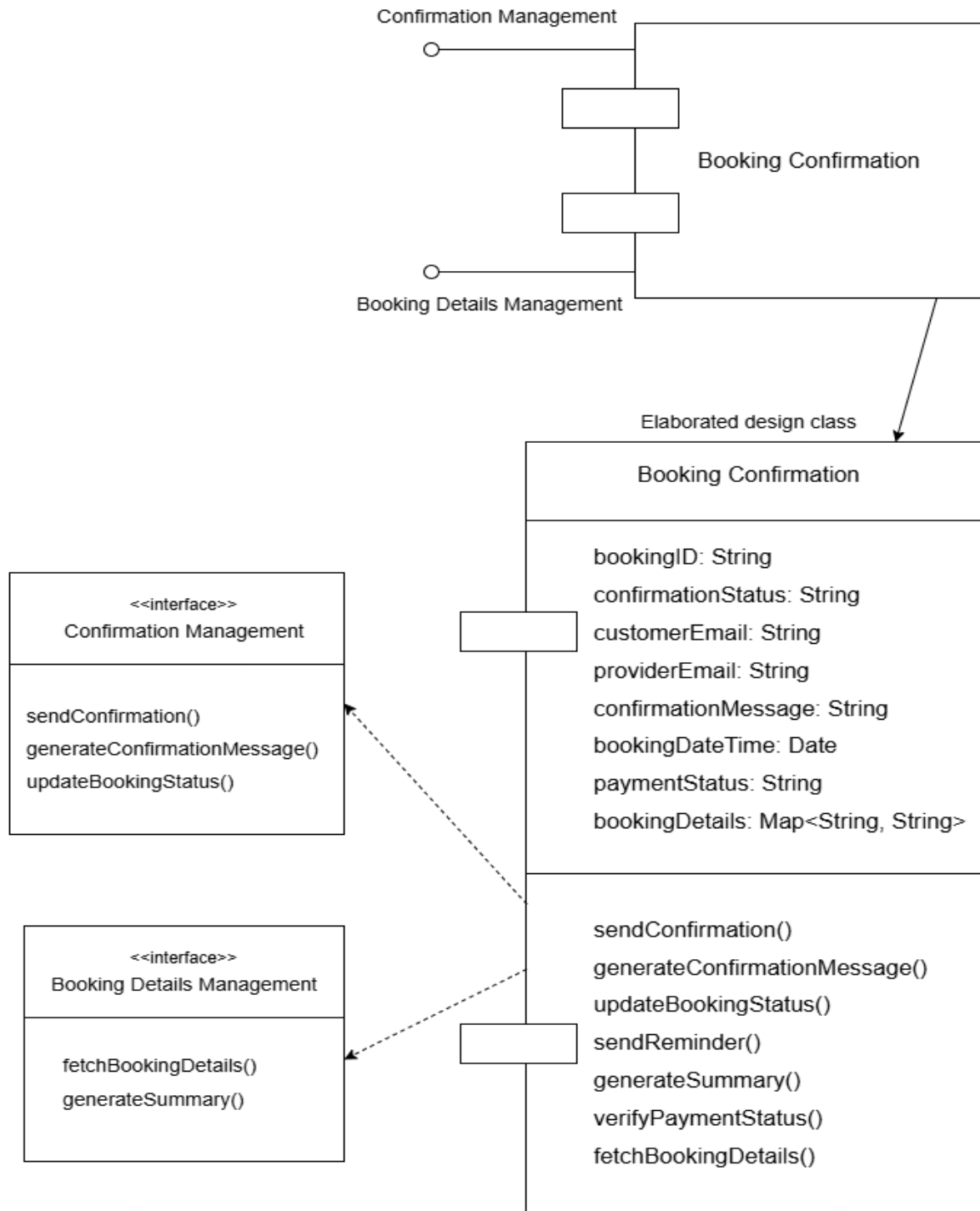


Figure 16:Elaboration of Booking Confirmation

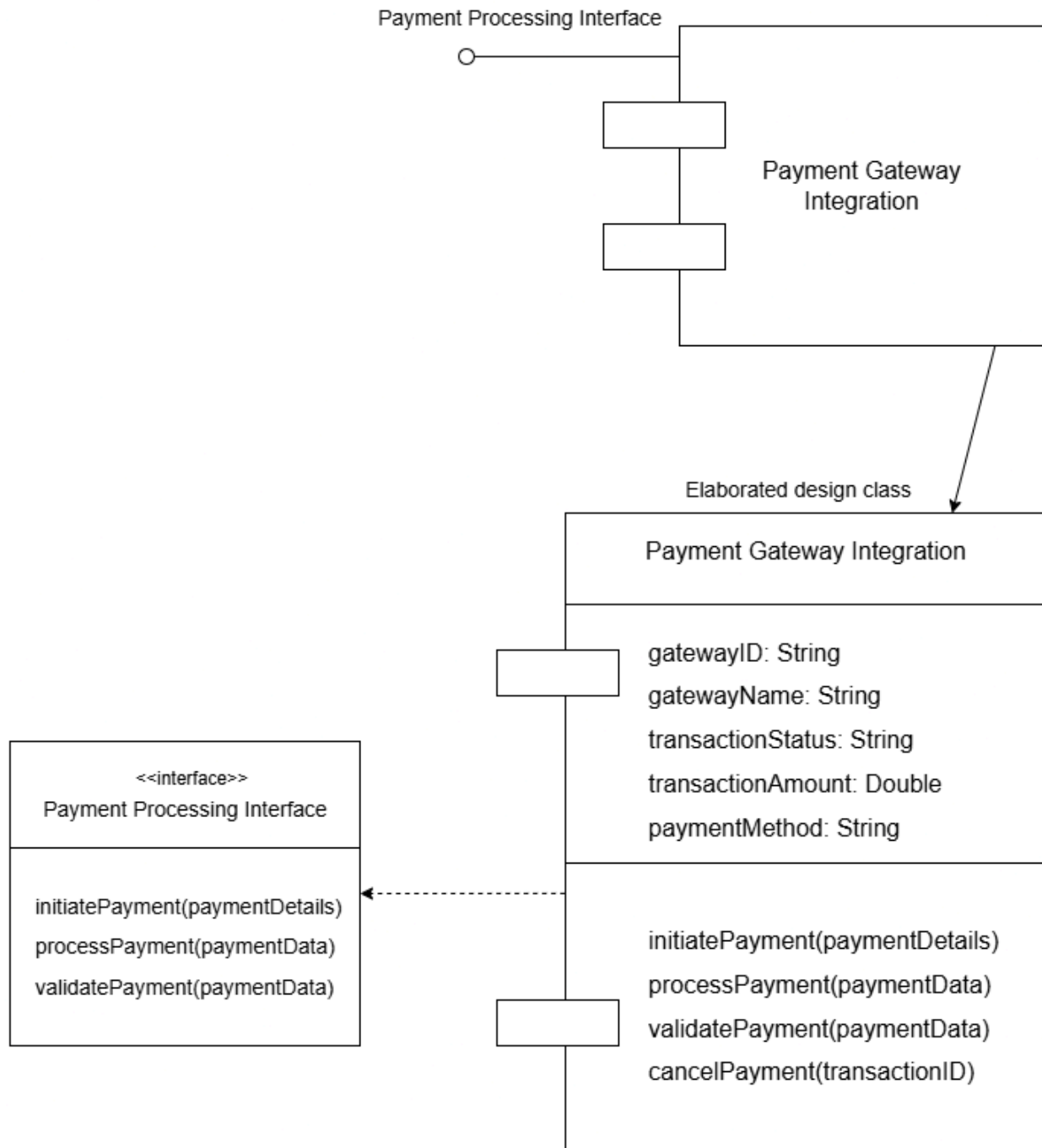


Figure 17: Elaboration of Payment Gateway Integration

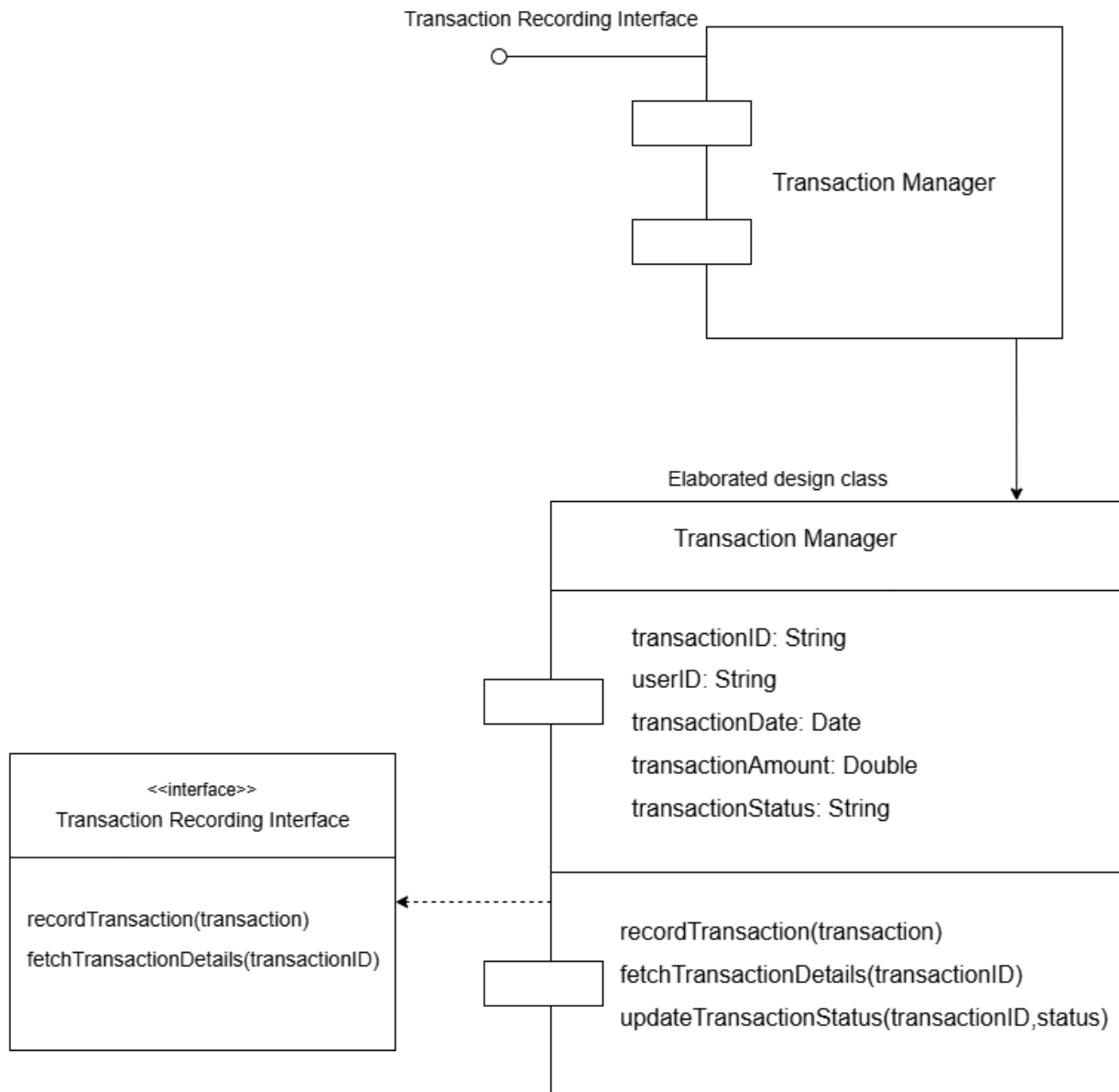


Figure 18: Elaboration of Transaction Manager

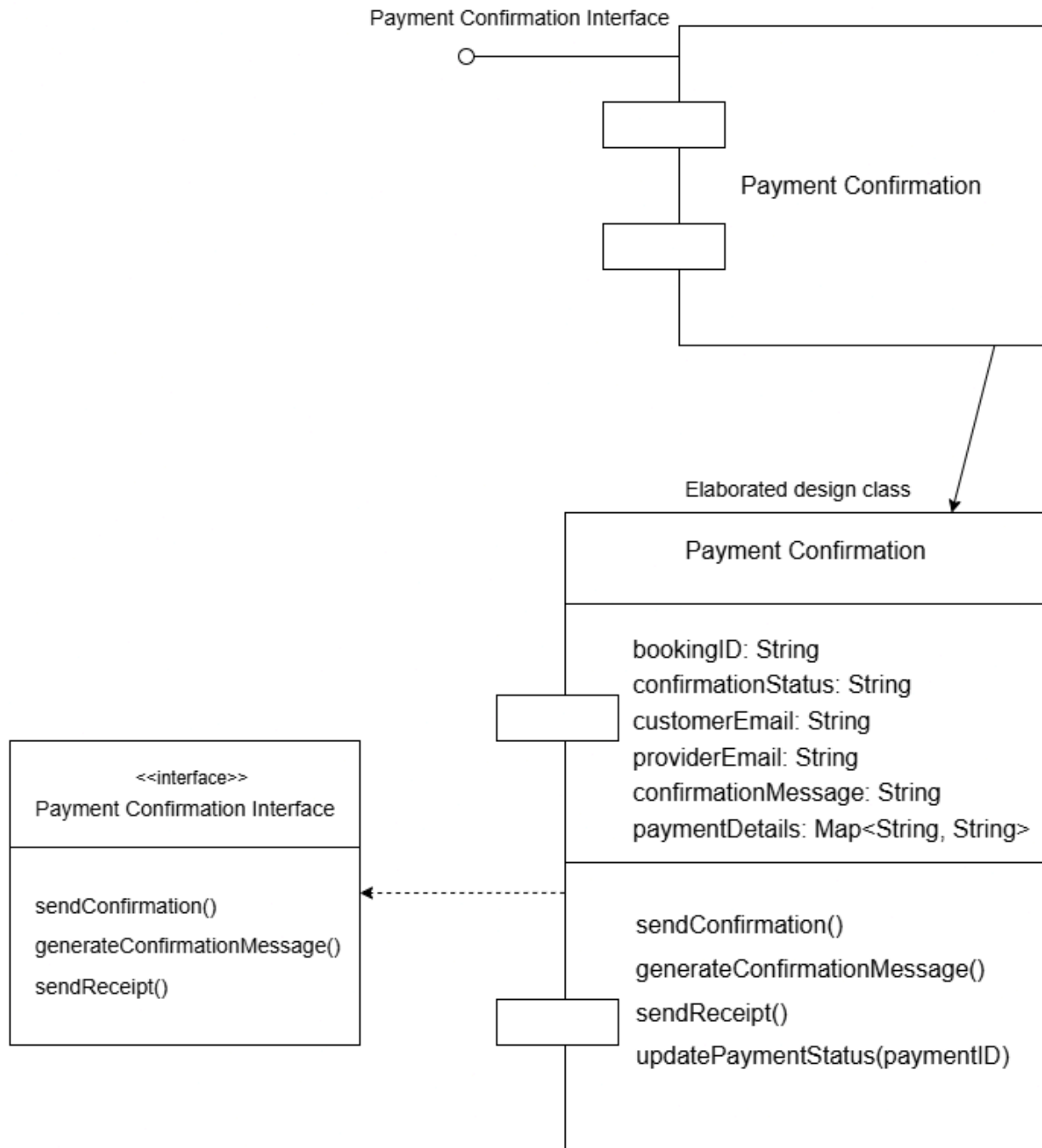


Figure 19: Elaboration of Payment Confirmation

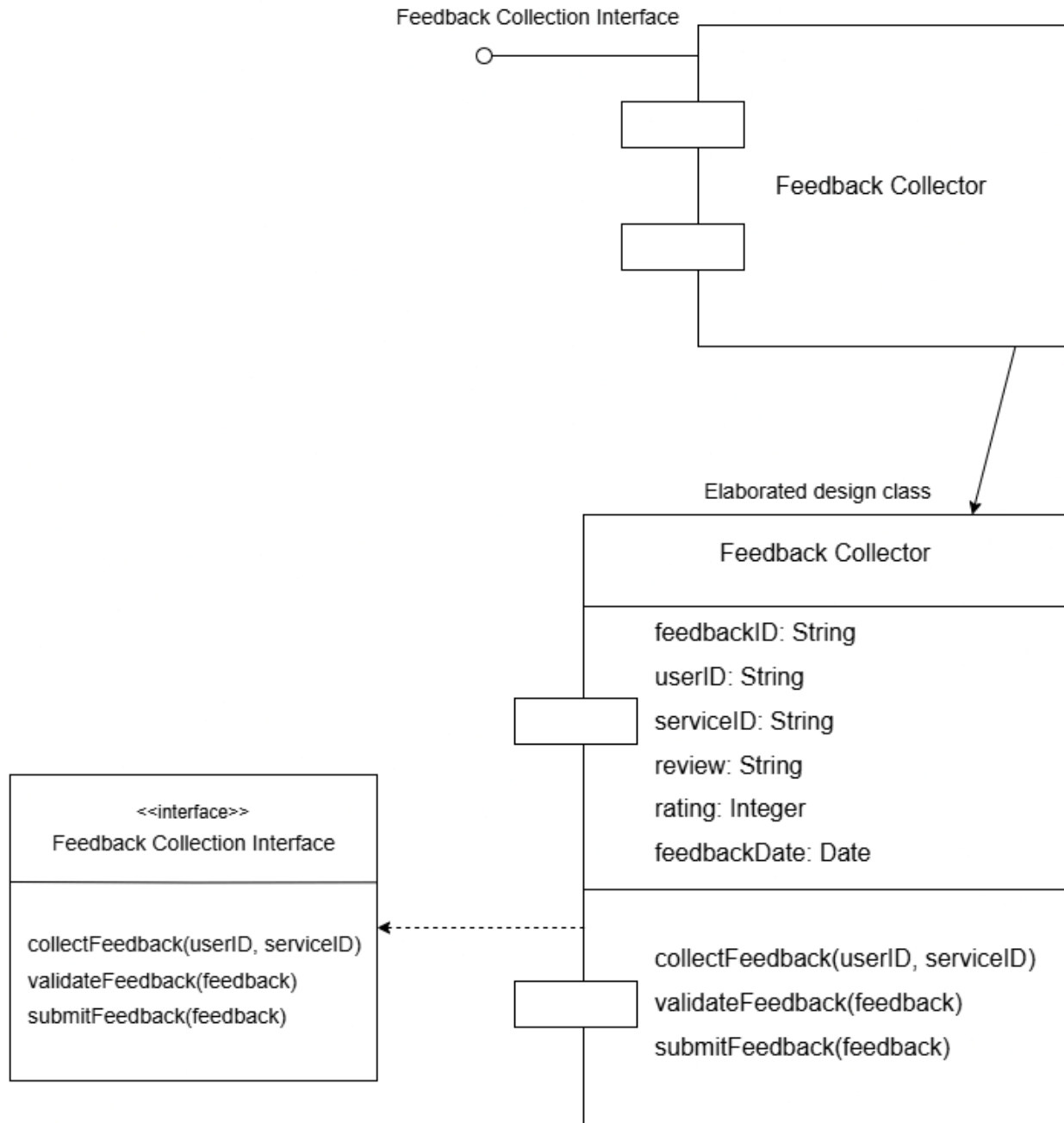


Figure 20: Elaboration of Feedback Collector

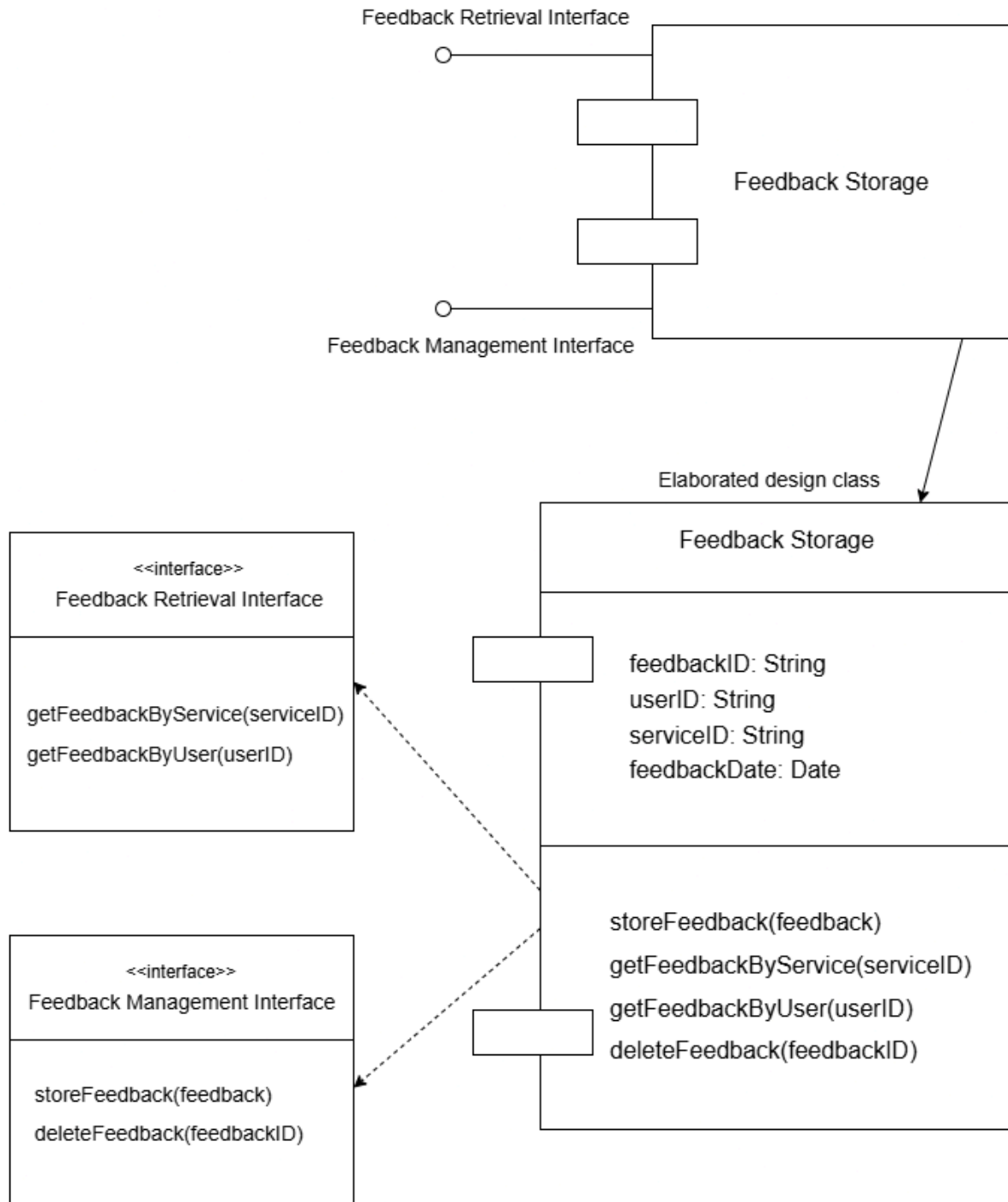


Figure 21: Elaboration of Feedback Storage

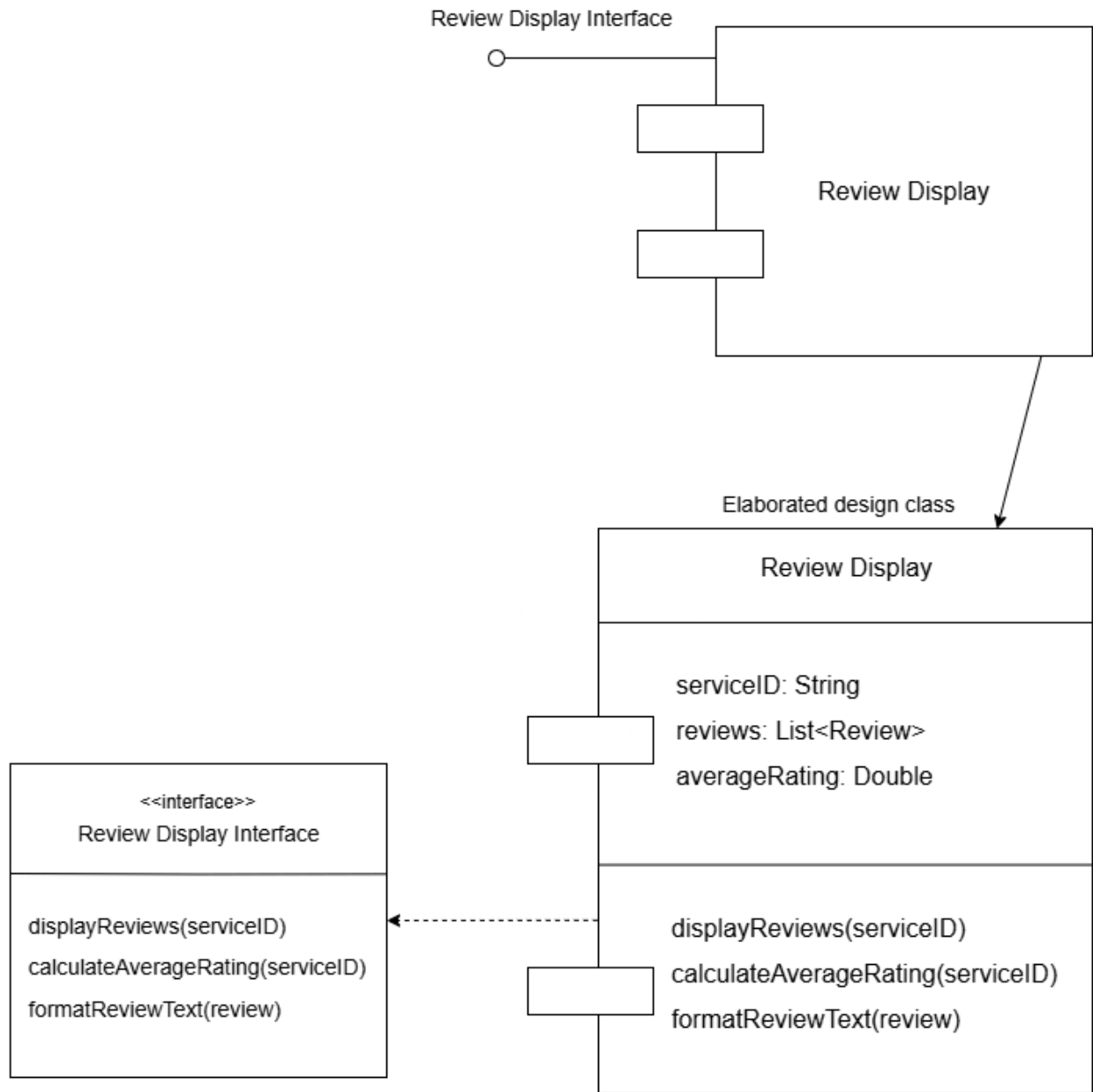


Figure 22: Elaboration of Review Display

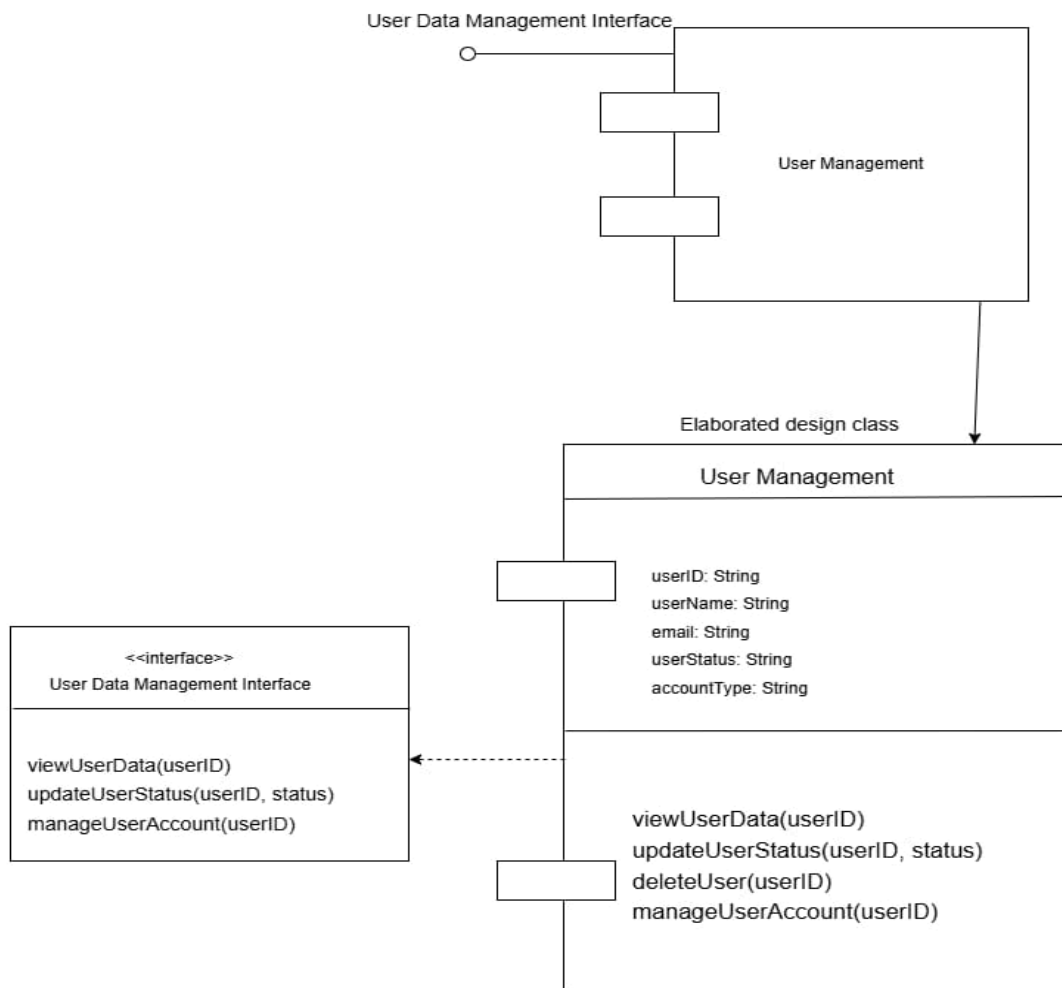


Figure 23: Elaboration of User Management

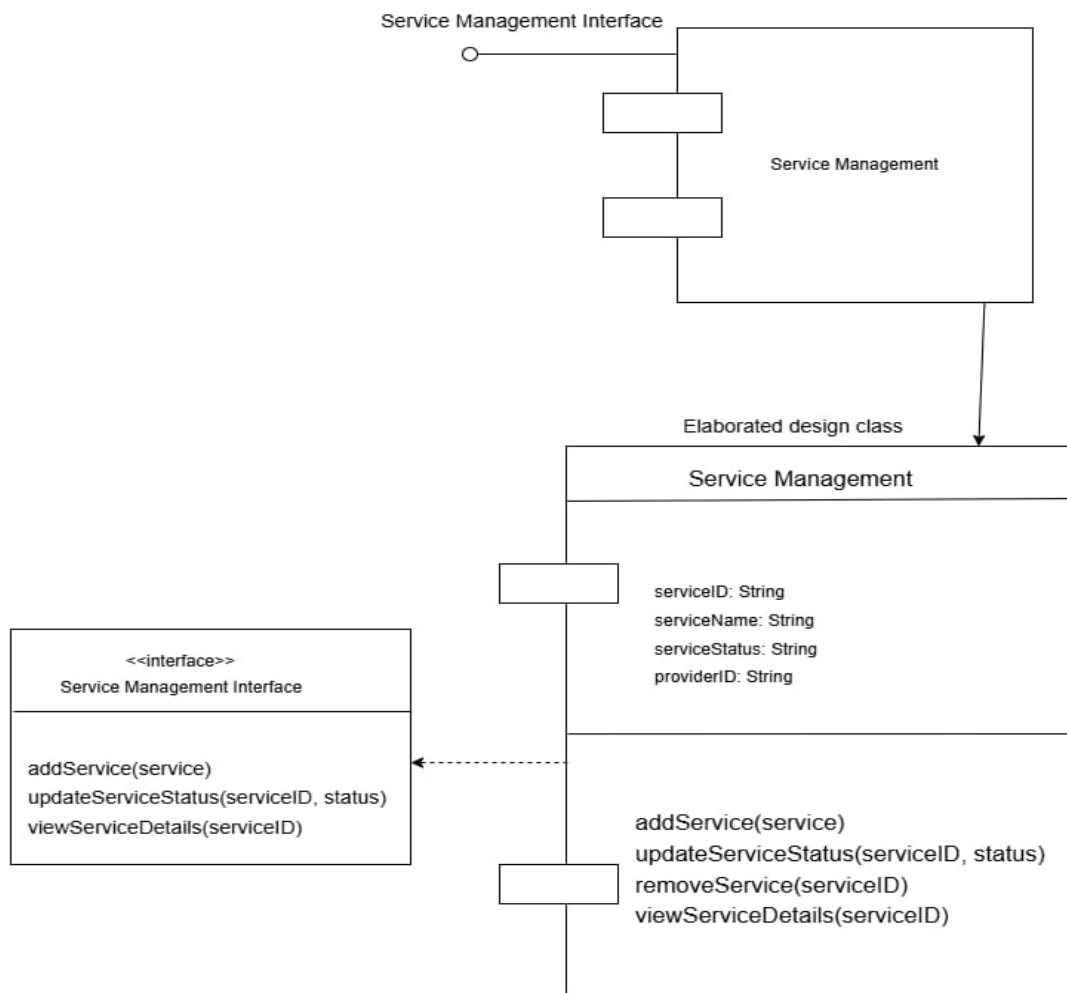


Figure 24: Elaboration of Service Management

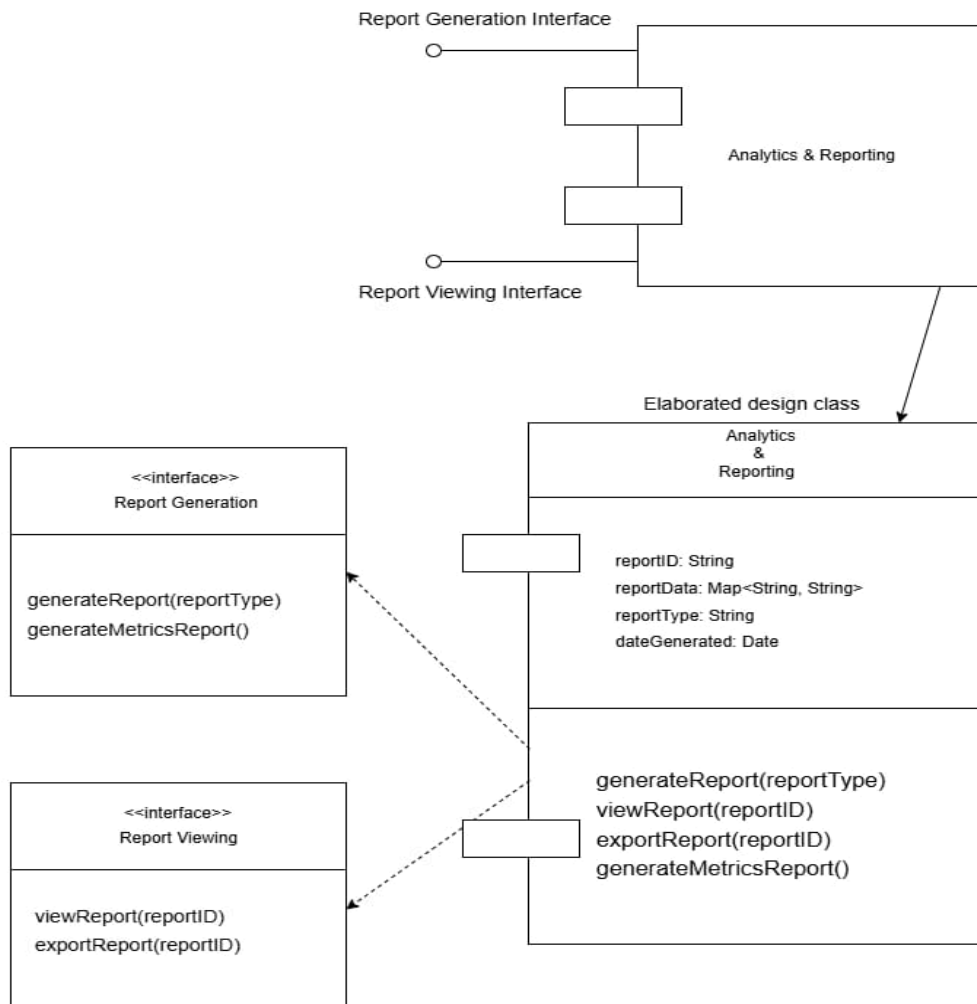


Figure 25: Elaboration of Analytics & Reporting

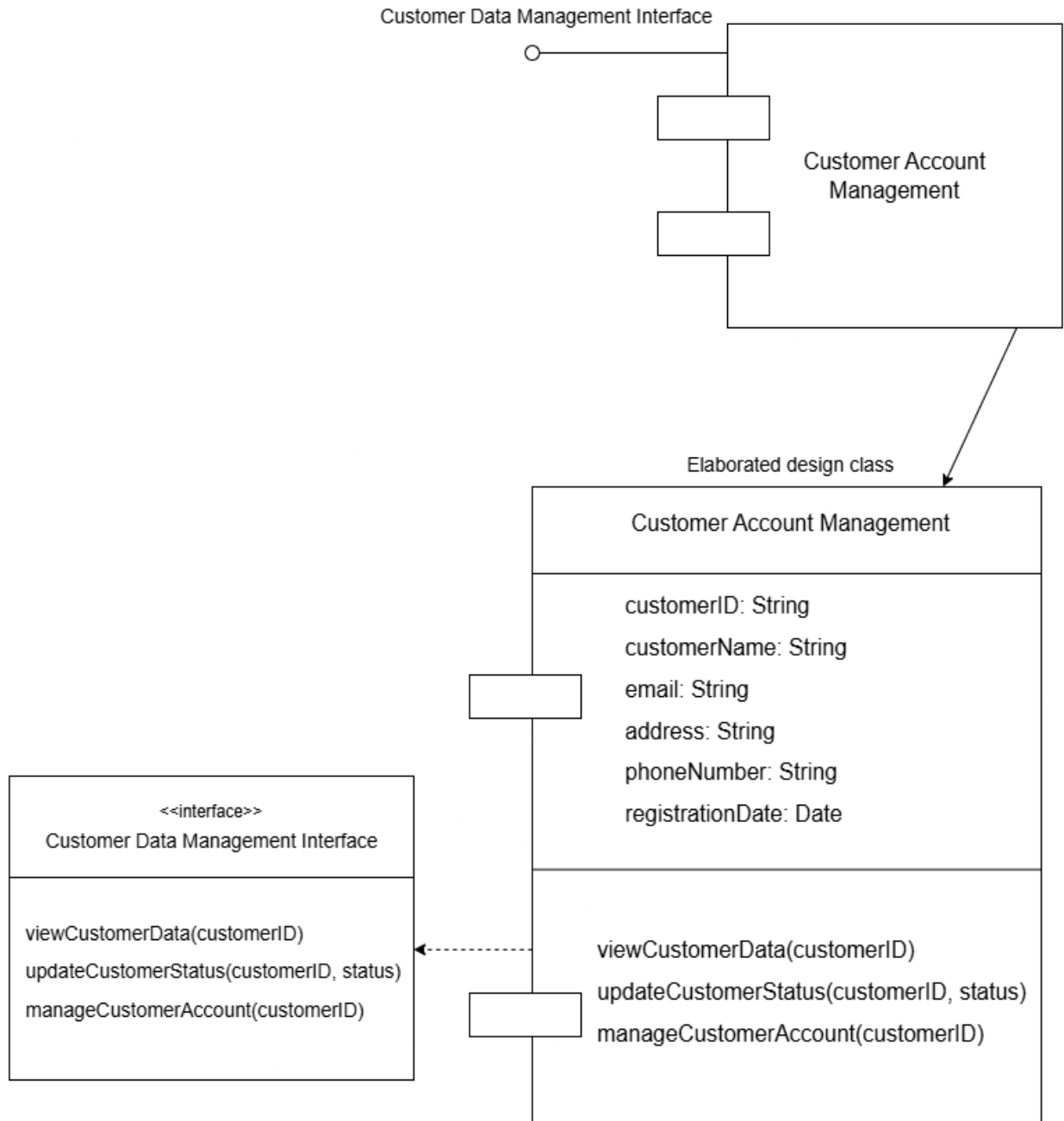


Figure 26: Elaboration of Customer Account Management

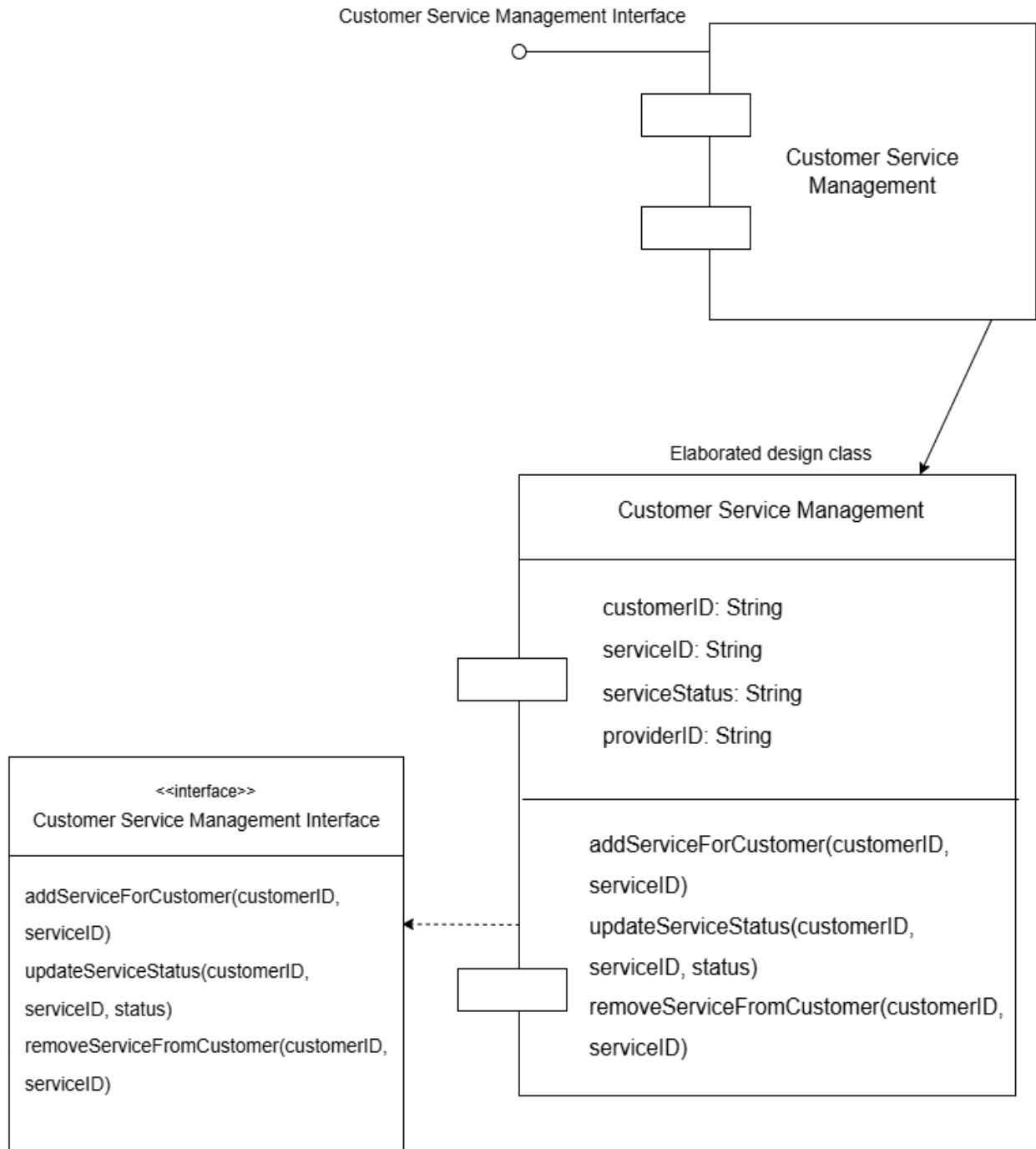


Figure 27: Elaboration of Customer Service Management

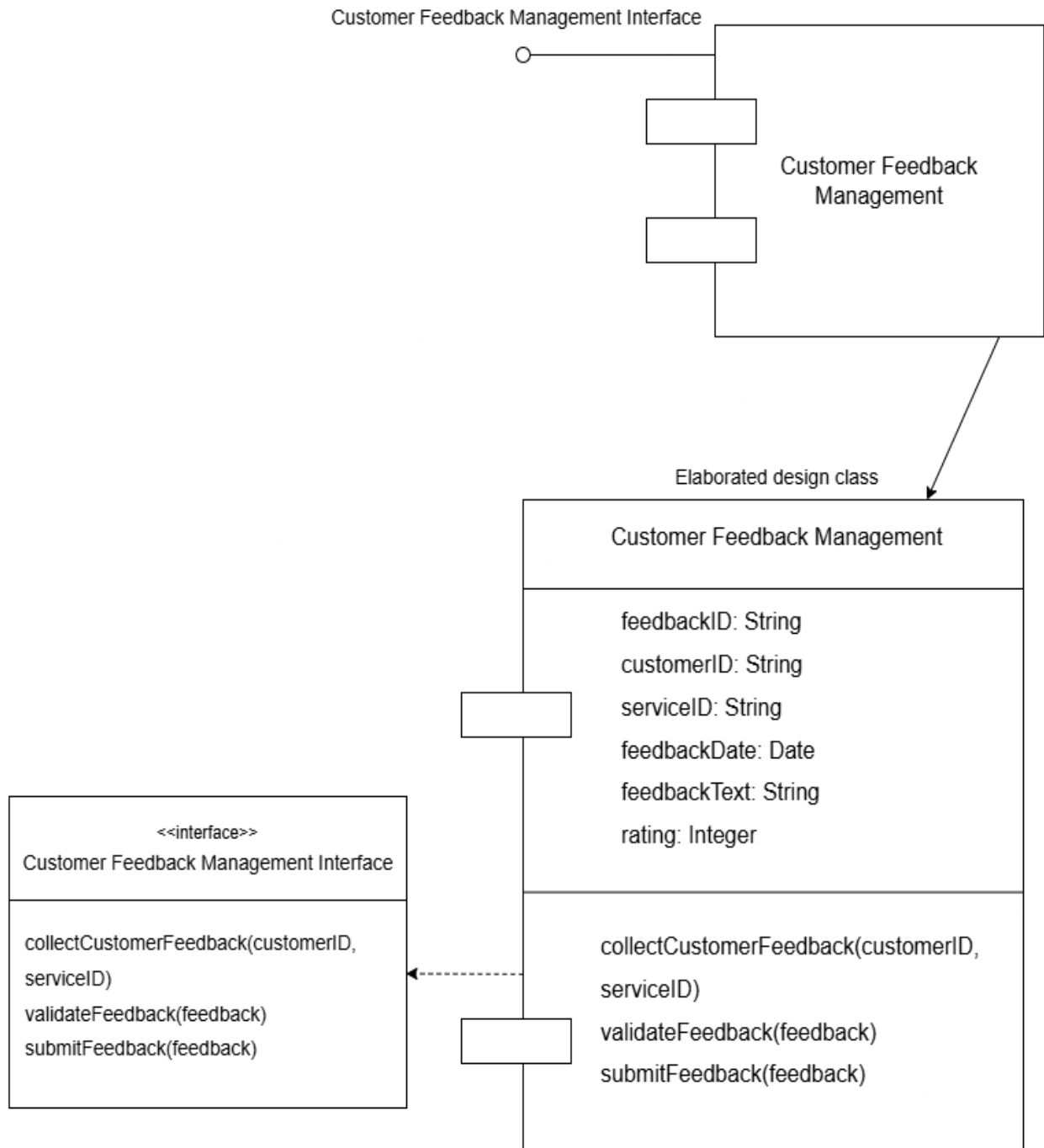


Figure 28: Elaboration of Customer Feedback Management

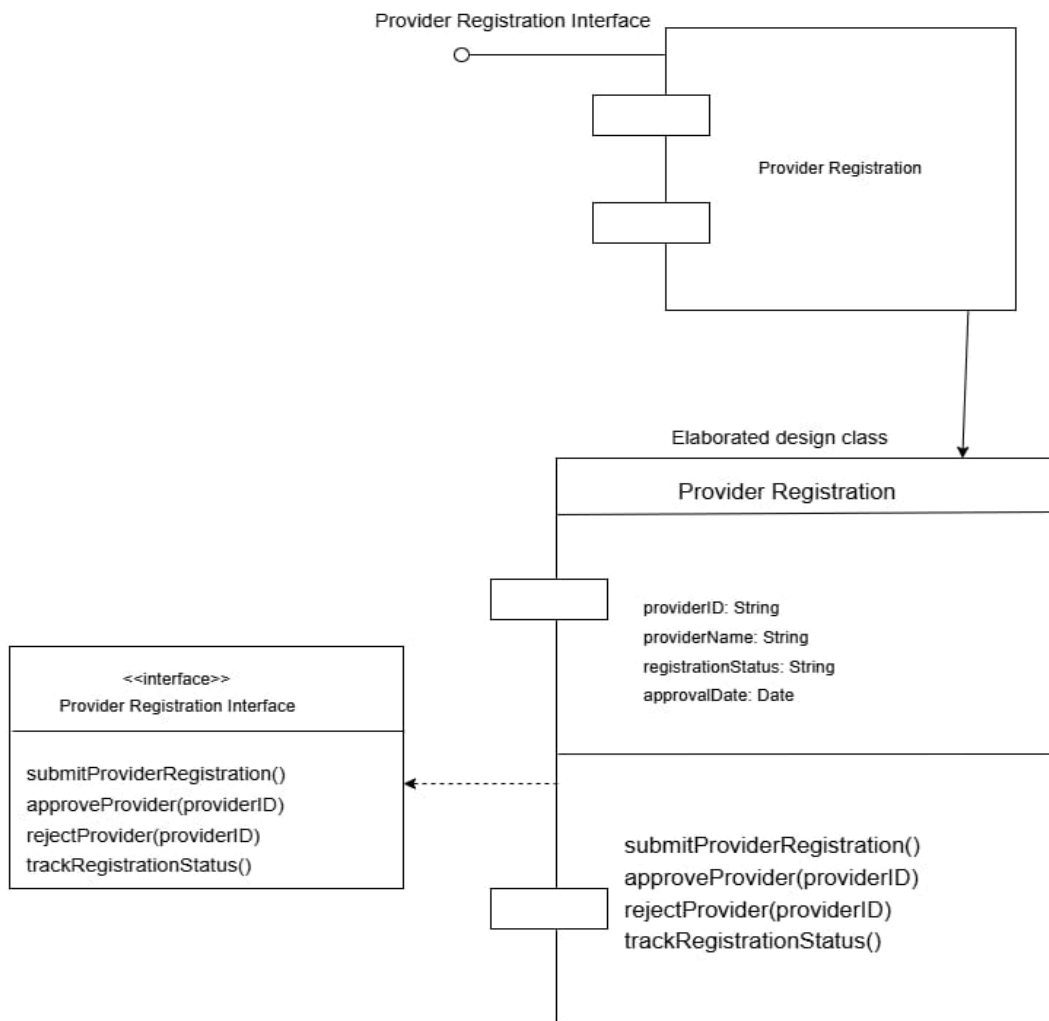


Figure 29: Elaboration of Provider Registration

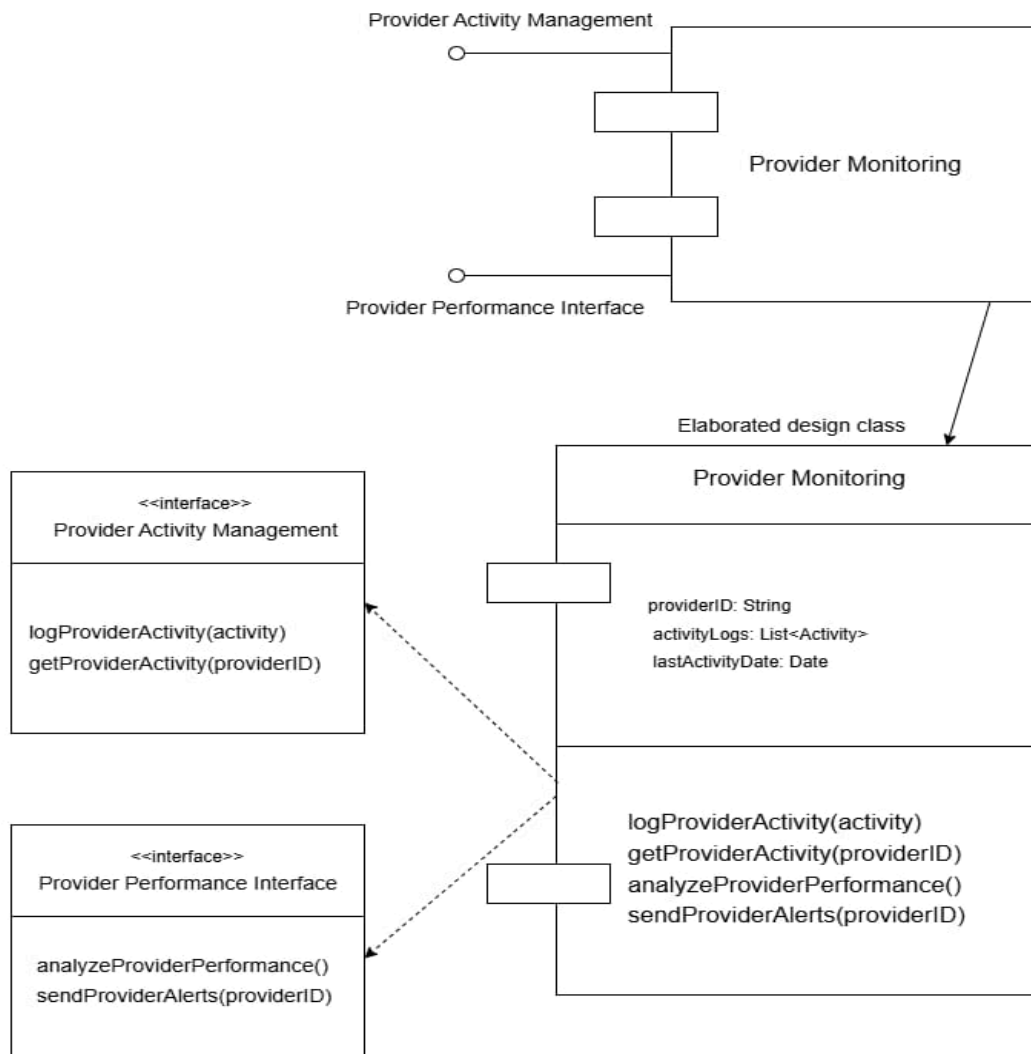


Figure 30: Elaboration of Provider Monitoring

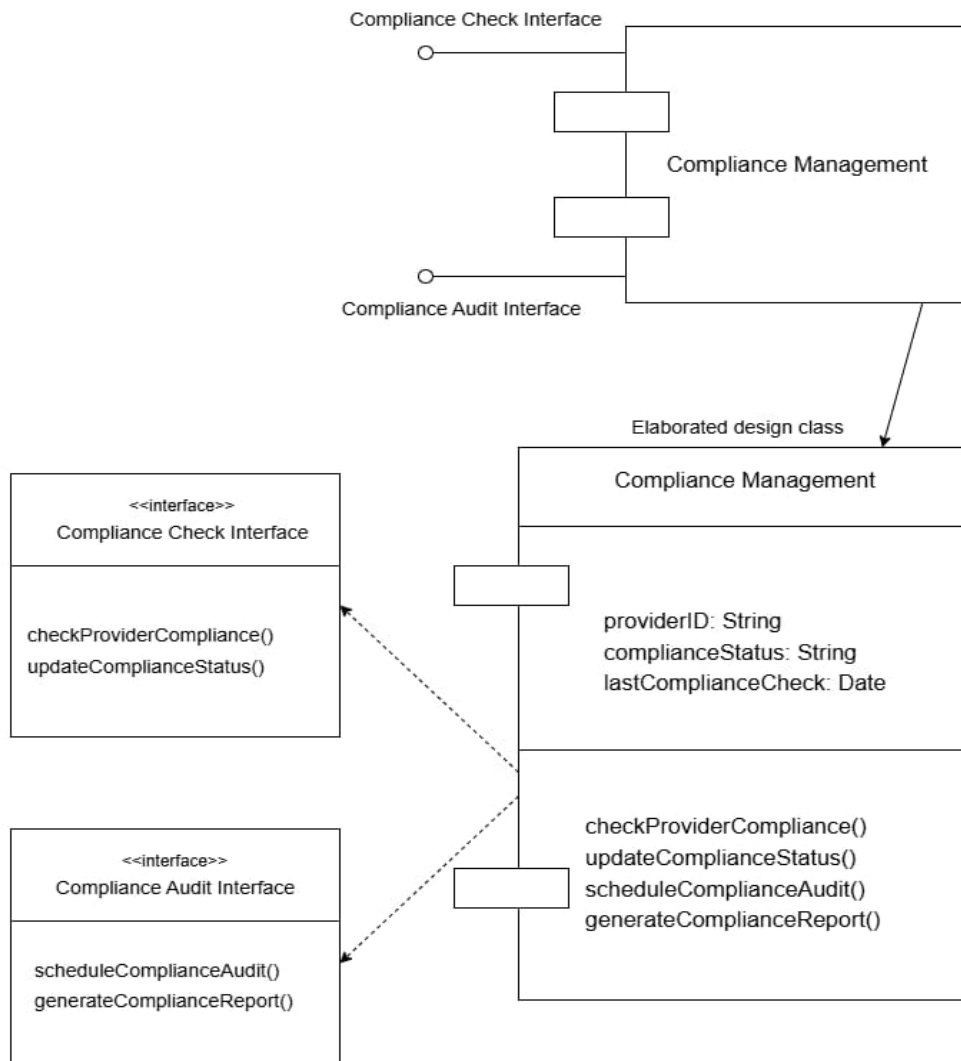


Figure 31: Elaboration of Compliance Management

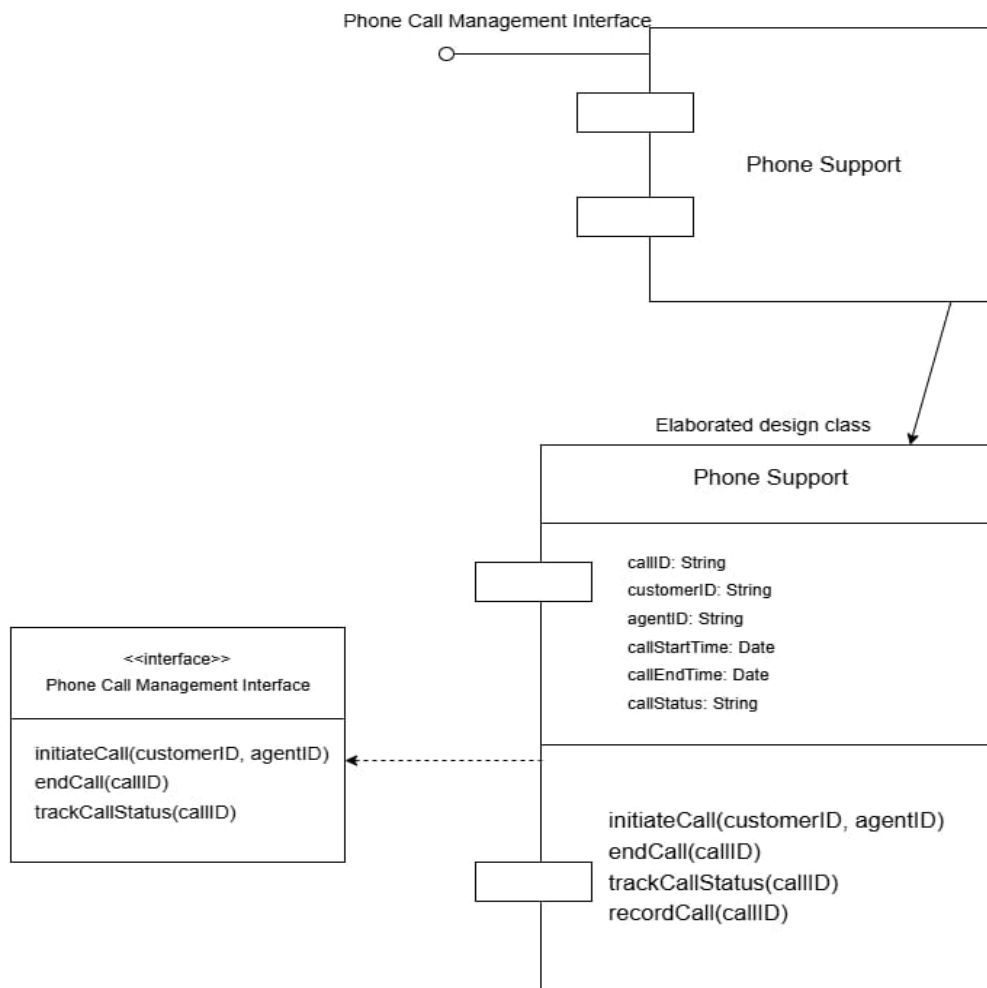


Figure 32: Elaboration of Phone Support

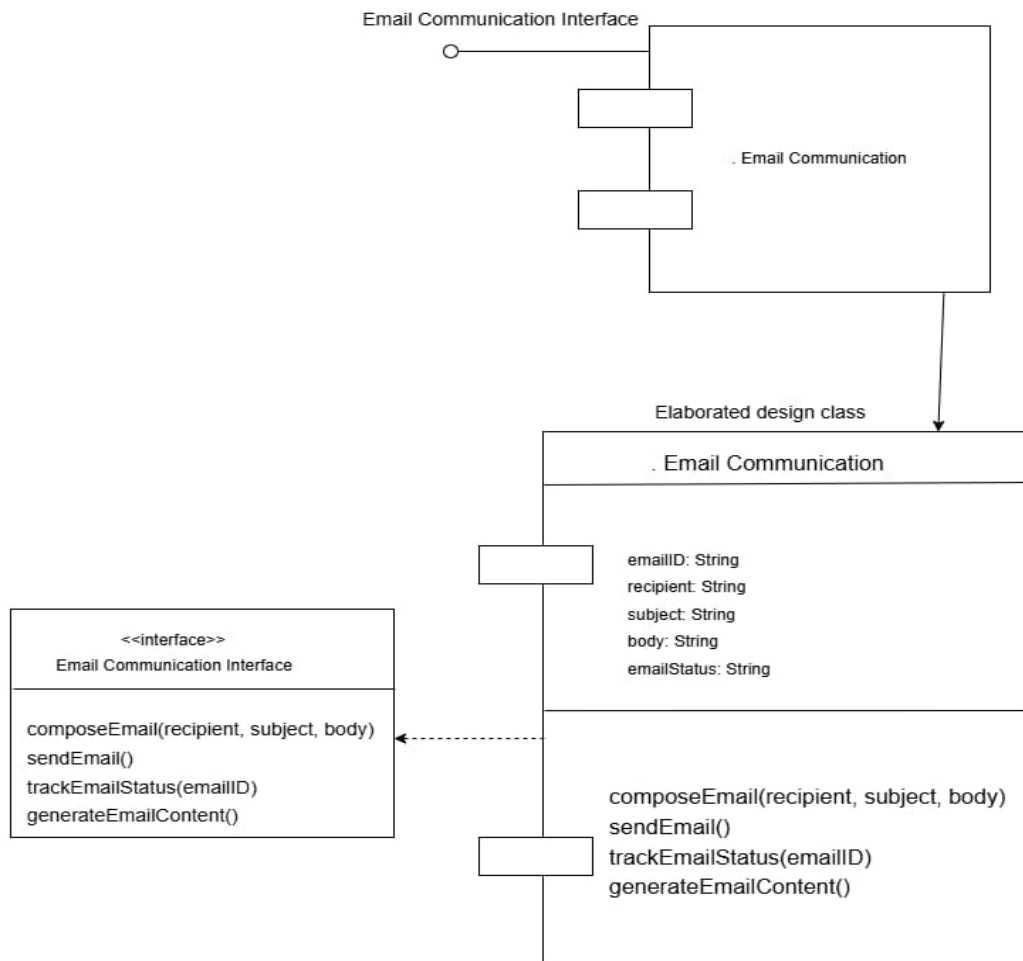


Figure 33: Elaboration of Email Communication

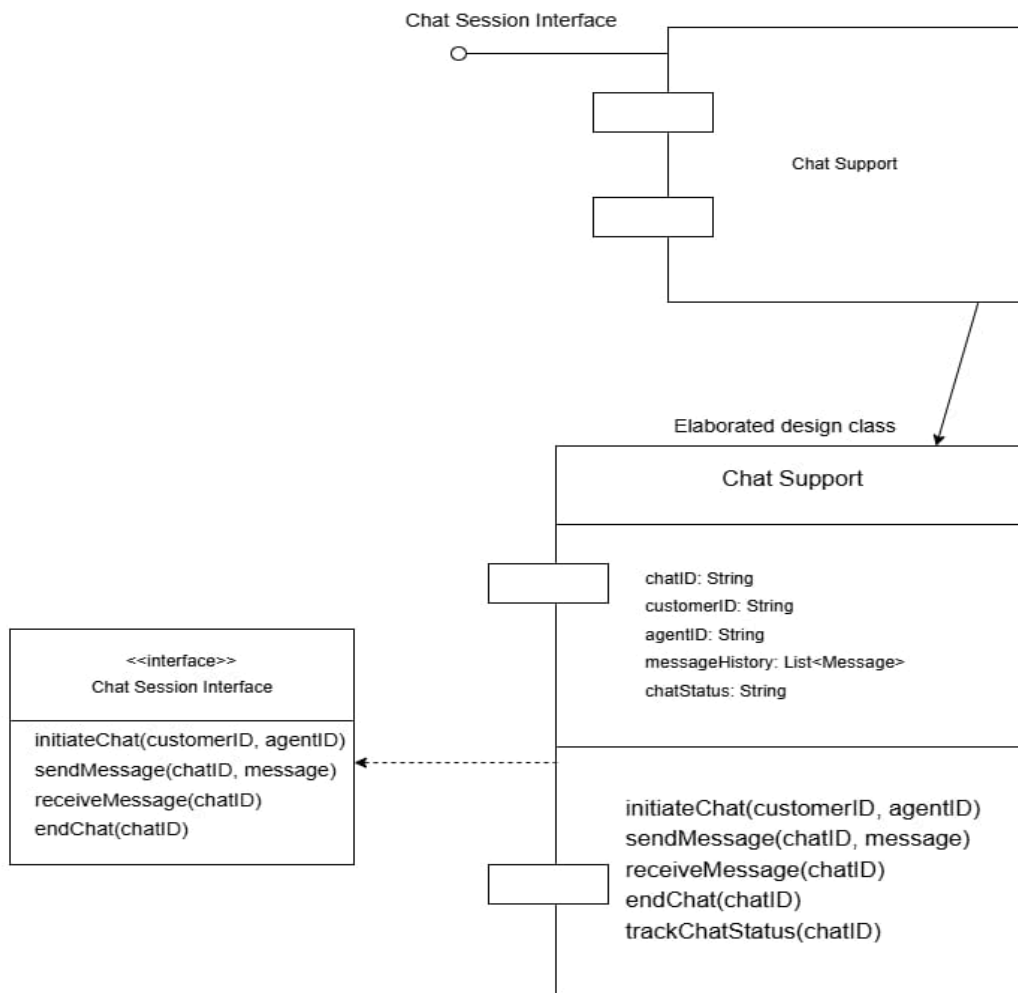


Figure 34: Elaboration of Chat Support

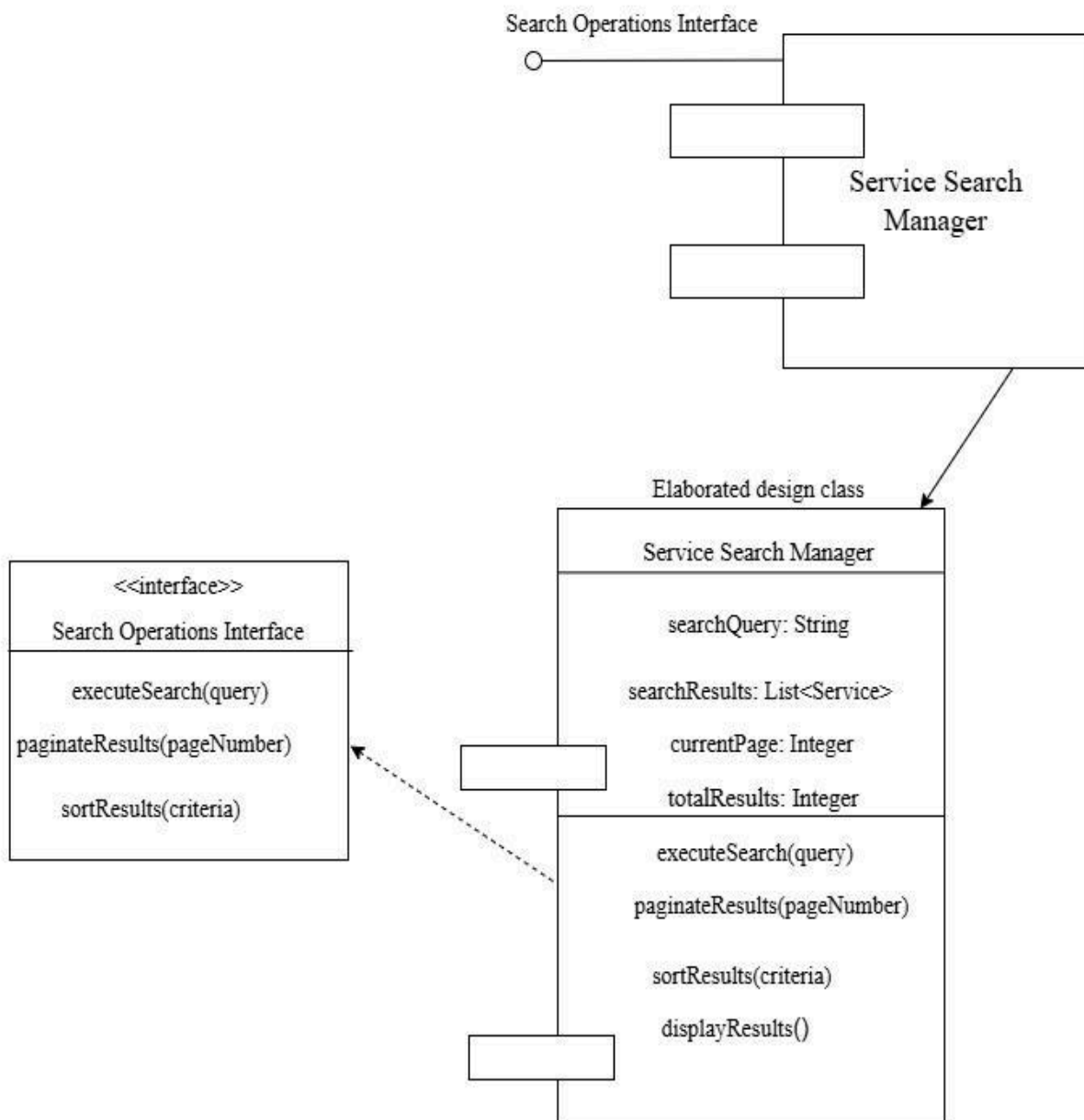


Figure 35: Elaboration of Service Search Manager

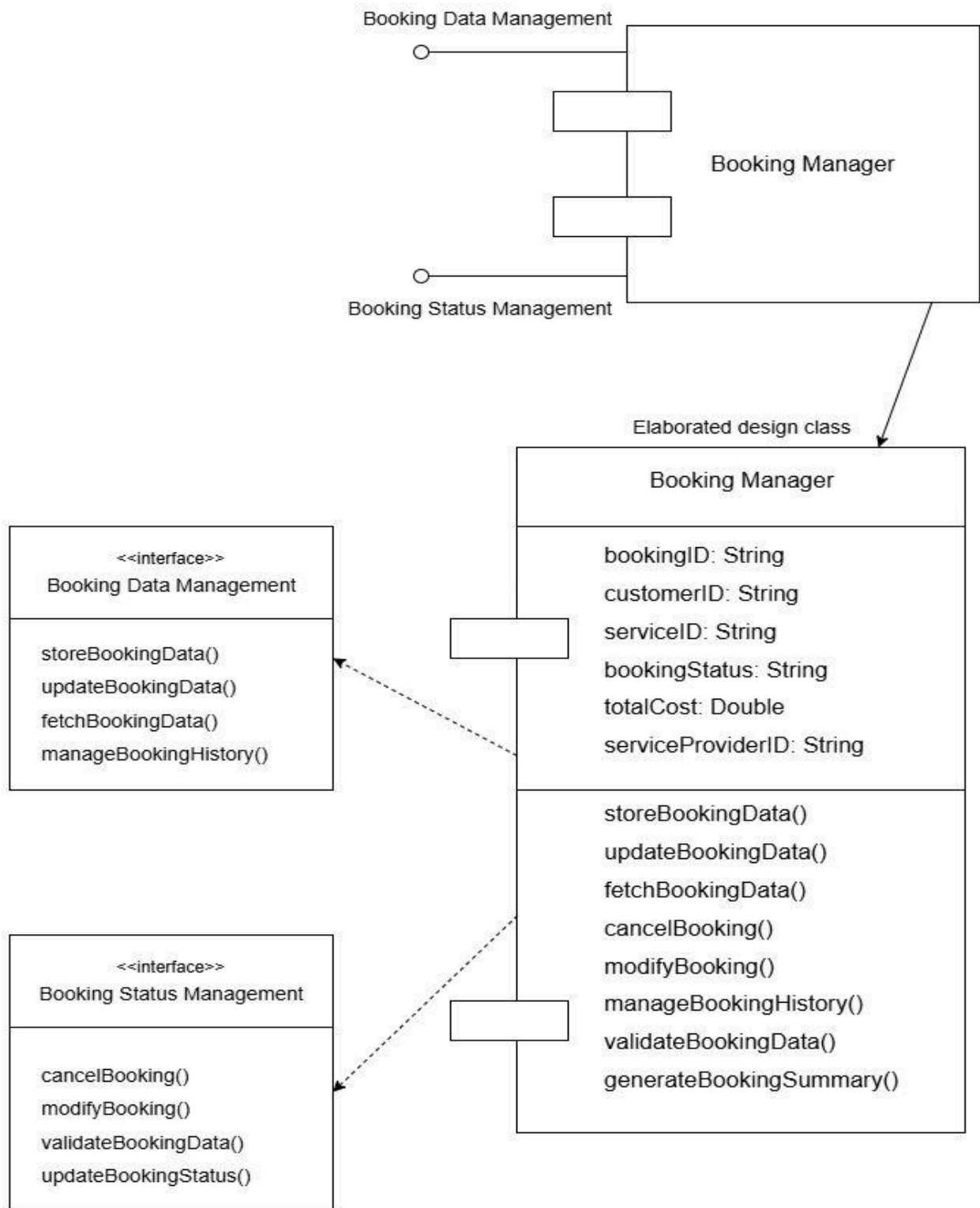


Figure 36: Elaboration of Booking Manager

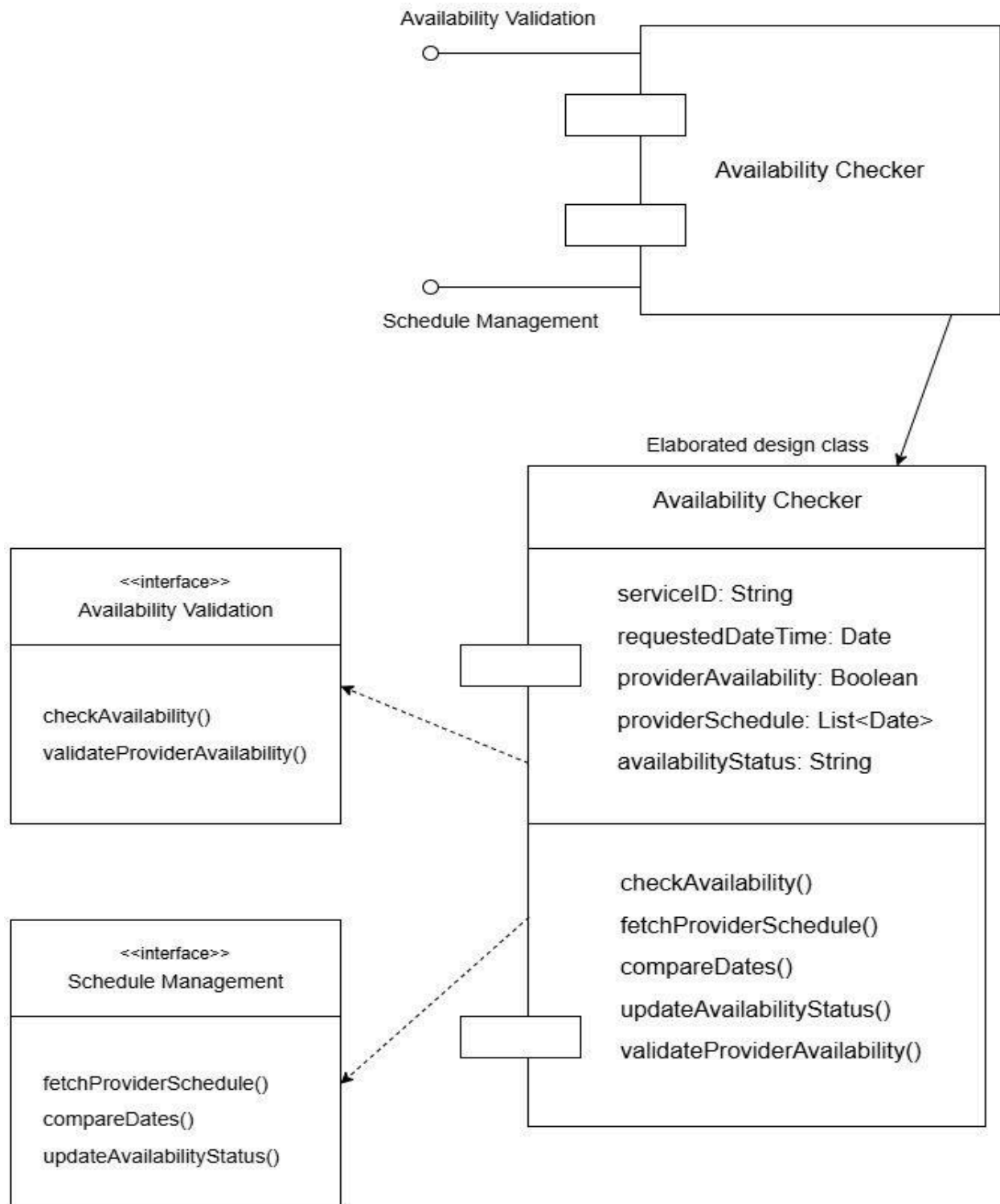


Figure 37: Elaboration of Availability Checker

3.2 Class Diagram

The Class Diagram represents the major software classes derived from the elaborated components. It shows the relationships between entities like Users, Services, Bookings, Payments, and Admin management. It helps to clearly define object-oriented structures, inheritance, associations, and dependencies needed for LocalLinker's development.

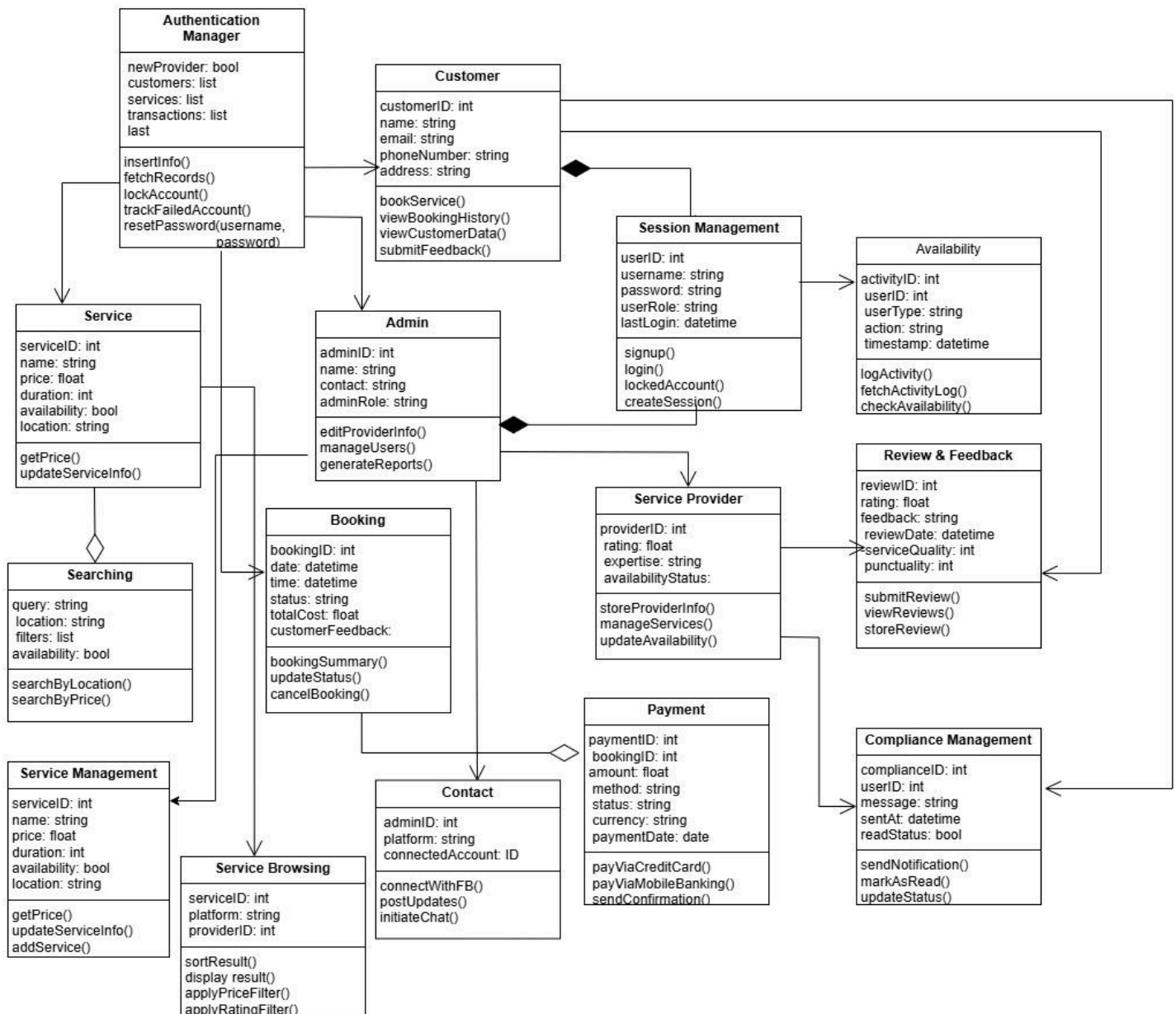


Figure 38:Class Diagram For LocalLinker

3.3 Database Design (ER Diagram, Tables, and Relationships)

The Database Design outlines the structure of the backend database using an Entity-Relationship (ER) diagram. It identifies the major tables (such as Users, Services, Bookings, Payments, Reviews), primary keys, foreign keys, and relationships among entities. A well-designed database ensures efficient data retrieval, integrity, scalability, and security for the LocalLinker platform.

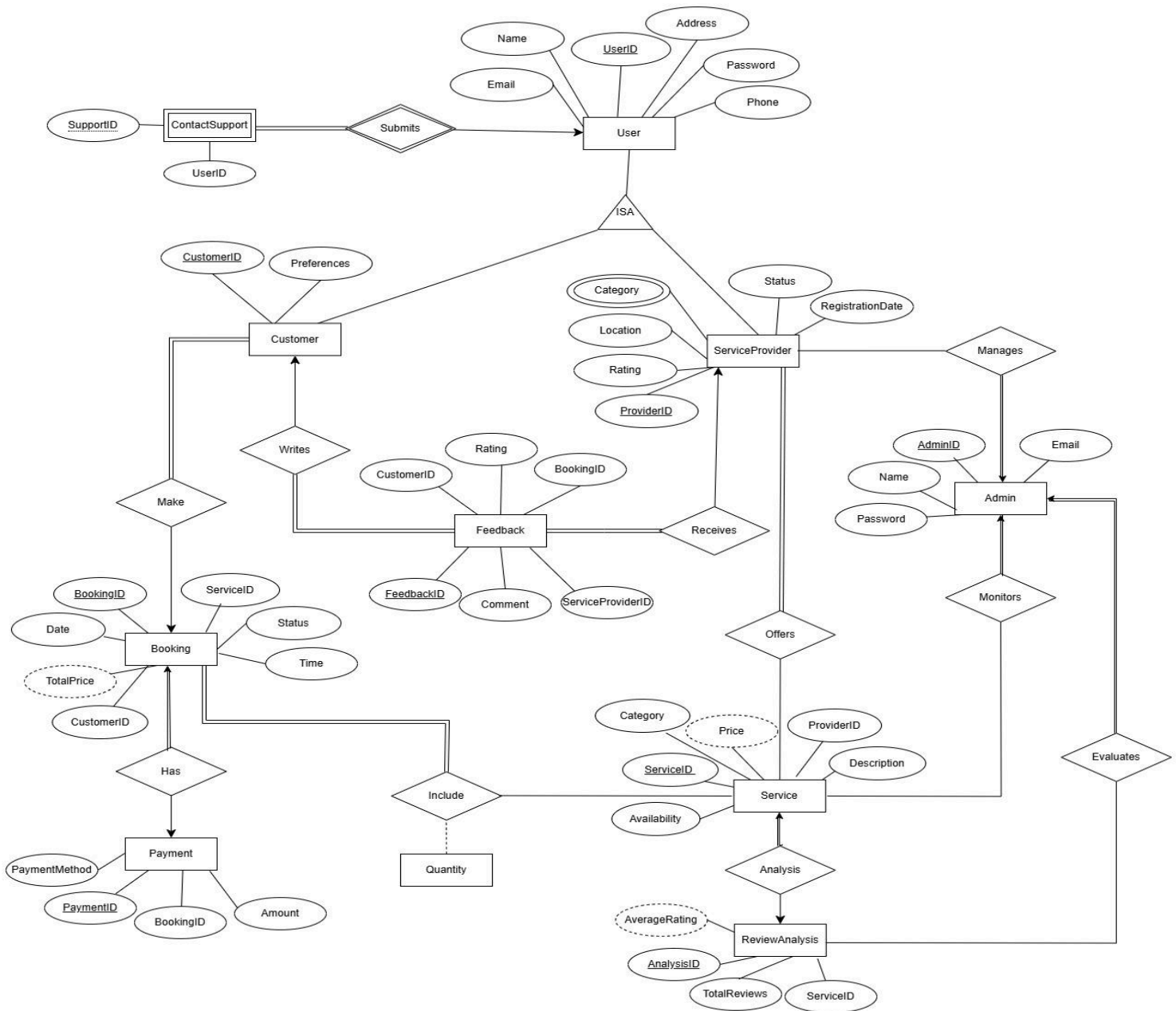


Figure 39: ER Diagram For LocalLinker

4. User Interface Design

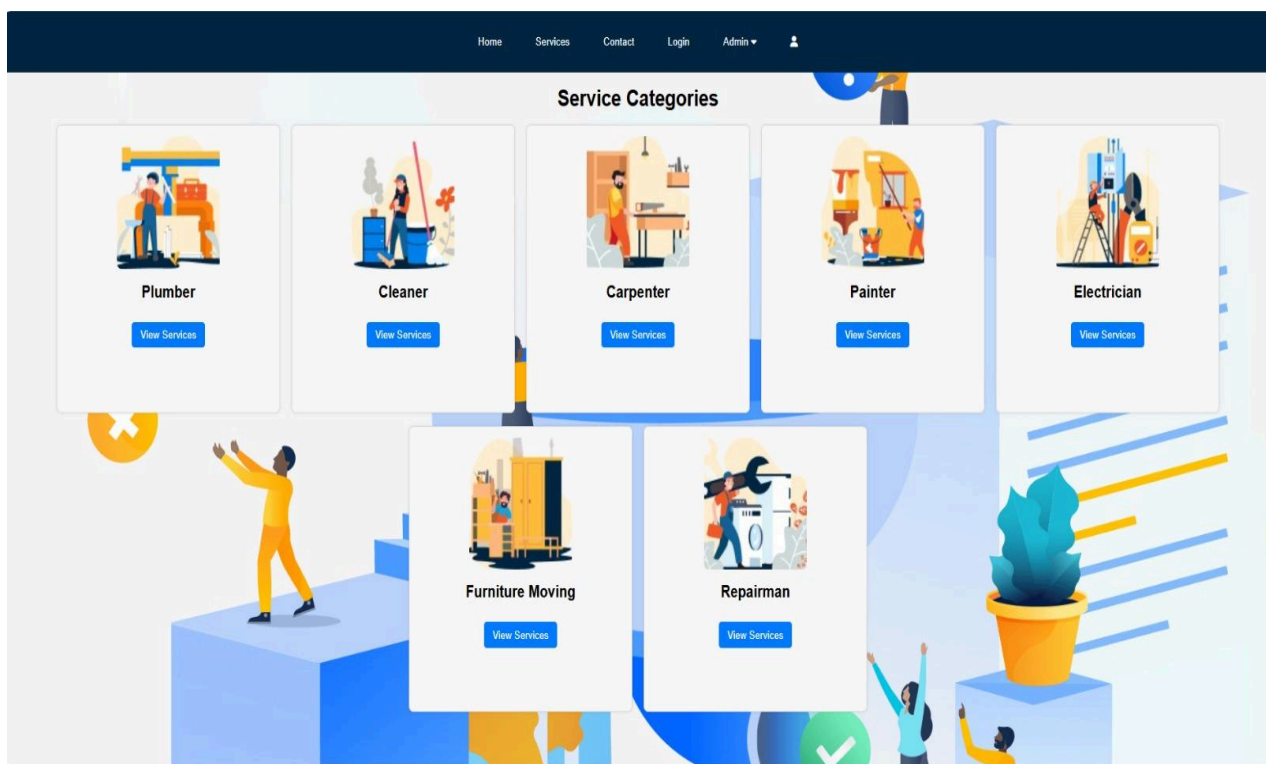
This section describes the visual layout and interaction flow of the LocalLinker system from the user's perspective. It focuses on how the platform's screens are designed to offer an intuitive, user-friendly, and responsive experience. Wireframes and navigation flows are used to represent major user actions and the transitions between different screens, ensuring that the platform is easy to use and visually consistent.

4.1UI Wireframes/Mockups (Depicting key screens and interactions)

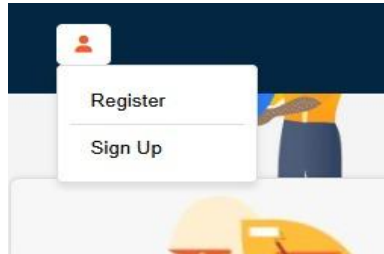
Wireframes and mockups illustrate the basic structure of key screens in the LocalLinker platform, such as the homepage, authentication pages, service browsing, booking process, and admin dashboard. These wireframes serve as a visual guide for developers, designers, and stakeholders to understand the user journey and refine the interface before full implementation.

These are our LocalLinker Project's Wireframes:

❖ Homepage of LocalLinker Website



❖ For Authentication For Customer And Service Provider



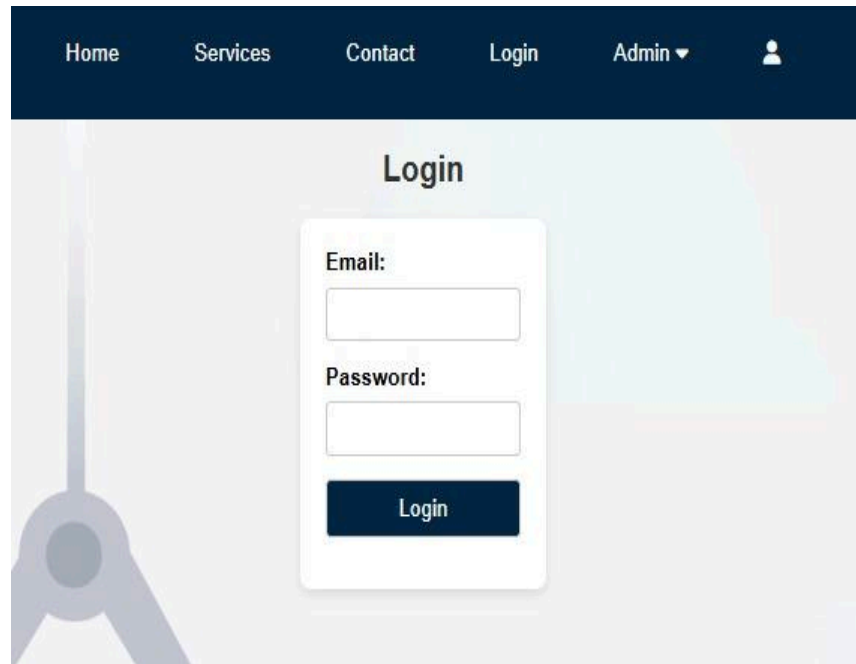
After Clicking SignUp As Customer
This Form Will Display->

A screenshot of a web application's 'Sign Up' form. The form is titled 'Sign Up' in bold. It contains several input fields: 'Name:', 'Email:', 'Phone:', 'Password:', 'Street:', 'Zip Code:', 'State:', and 'City:'. Each field has a corresponding label and a text input box. At the bottom of the form is a dark blue button labeled 'Sign Up'. The background of the page shows a stylized robot arm and a yellow circle with a white plus sign. The footer includes the copyright notice '© 2025 Local Service Management. All rights reserved.' and links for 'About Us', 'Contact', and 'Privacy Policy'.

After Clicking Register As Service Provider
This Form Will Display->

A screenshot of a web application's 'Register as a Service Provider' form. The form is titled 'Register as a Service Provider' in bold. It contains several input fields: 'Provider Name:', 'Category:' (with a dropdown menu), 'Price (USD):', 'Service Description:', 'Provider Location:', 'Gender:' (with a dropdown menu), 'Email:', 'Password:', 'Phone:', 'Present Address:', and 'Permanent Address:'. Each field has a corresponding label and a text input box. At the bottom of the form is a dark blue button labeled 'Register'. The background of the page shows a stylized robot arm and a yellow circle with a white plus sign. The footer includes the copyright notice '© 2025 Local Service Management. All rights reserved.' and links for 'About Us', 'Contact', and 'Privacy Policy'.

❖ After Completing Authentication Login Page



The screenshot displays a web application's login page. At the top, a dark blue navigation bar contains the links 'Home', 'Services', 'Contact', 'Login', and 'Admin' with a dropdown arrow, followed by a user profile icon. The main content area has a light gray background with a faint illustration of a person. Centered on the page is a white login form with a dark blue border. The form is titled 'Login' in a large, bold, dark blue font. It contains two input fields: 'Email:' and 'Password:', each with a corresponding text box. Below these fields is a dark blue 'Login' button with white text. The background of the page features a light blue and white abstract design.

Customer And Service Providers Log in Interface

❖ Contact Page

[Home](#) [Services](#) [Contact](#) [Login](#) [Admin](#) 

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